EasyVisa Business Report

Robust Data Modeling with Ensembling and Tuning Techniques To analyse and Predict Visa Approvals

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I. Problem Context:

While US businesses face a strong demand for human resources, finding and retaining the right talent remains a critical competitive challenge. To fill these gaps, companies seek qualified professionals both domestically and internationally. Under the Immigration and Nationality Act (INA), foreign professionals can work in the US temporarily or permanently, provided that employers adhere to regulations designed to protect the wages and working conditions of the domestic workforce. This process is overseen by the Office of Foreign Labor Certification (OFLC), which approves labor certifications only when employers prove that qualified US workers are unavailable and that foreign hires will be paid prevailing regional wages.

II. Objective:

As a data scientist at EasyVisa, the goal is to leverage the historical data provided by the OFLC to build a predictive classification Machine Learning model that addresses two primary goals:

- **Streamline Visa Approvals:** Automate and facilitate the screening process by pre-shortlisting applicants who have a high probability of visa approval, thereby reducing the manual burden on reviewers.
- **Identify Key Decision Drivers:** Analyze and recommend the specific profile characteristics (such as education, job type, or wage rate) that most significantly influence whether a visa application is certified or denied.

III. Problem Definition:

US businesses face a continuous challenge in identifying and attracting the right talent to remain competitive, often looking both locally and abroad to fill critical skill gaps. Under the Immigration and Nationality Act (INA), foreign workers can be hired on a temporary or permanent basis, provided domestic labor standards, wages, and working conditions are protected.

The Office of Foreign Labor Certification (OFLC) is responsible for processing these labor certification applications, ensuring foreign workers are only brought in when there is a genuine shortage of qualified US citizens. However, this workload has grown dramatically. For instance, in FY 2016, the OFLC processed 775,979 employer applications for nearly 1.7 million positions—representing a 9% increase from the previous year. Because the volume of applicants continues to rise annually, the manual review process for every single case has become incredibly tedious, inefficient, and difficult to scale.

IV. Exploratory Data Analysis

IV.1. Data Description:

The data contains the different attributes of customers' booking details. The detailed data dictionary is given below.

Data Dictionary:

- case_id: ID of each visa application

- continent: Information of continent the employee
- education_of_employee: Information of education of the employee
- has_job_experience: Does the employee have any job experience? Y= Yes; N = No
- requires_job_training: Does the employee require any job training? Y = Yes; N = No
- no_of_employees: Number of employees in the employer's company
- yr_of_estab: Year in which the employer's company was established
- region_of_employment: Information of foreign worker's intended
- region of employment in the US.
- prevailing_wage: Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- unit_of_wage: Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- full_time_position: Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
- case_status: Flag indicating if the Visa was certified or denied

IV.2. Data Quality Checks:

IV.2.1. Unique Values:

1. Unique values for 'continent': ['Asia' 'Africa' 'North America' 'Europe' 'South America' 'Oceania']
2. Unique values for 'education_of_employee': ['High School' "Master's" "Bachelor's" 'Doctorate']
3. Unique values for 'region_of_employment': ['West' 'Northeast' 'South' 'Midwest' 'Island']
4. Unique values for 'unit_of_wage': ['Hour' 'Year' 'Week' 'Month']
5. Unique values for 'case_status': ['Denied' 'Certified']

IV.2.2. Missing Values:

There are 25480 rows and 12 columns in the dataset. There are no missing values in data

IV.2.3. Duplicate Values:

There are no duplicate values in data

IV.3. Univariate Analysis:

IV.3.1. Univariate Analysis For Numerical Columns

IV.3.1.1. Univariate Analysis For No of Employees:

IV.3.1.1.1. Descriptive statistics for no_of_employees:

count	25480.000
mean	5667.043

std 22877.929
min -26.000
25% 1022.000
50% 2109.000
75% 3504.000
max 602069.000

Name: no_of_employees, dtype: float64

IV.3.1.1.2. Visual Representation for no_of_employees:

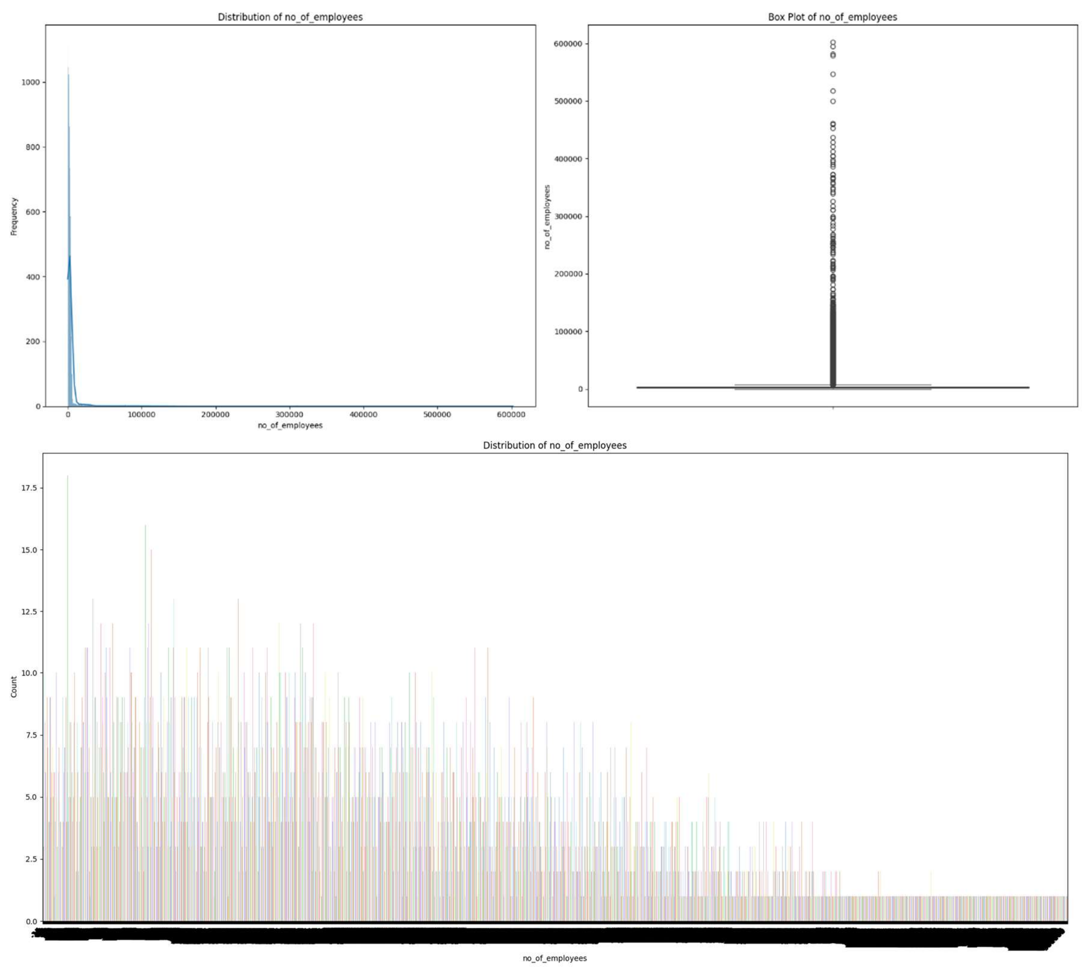


Figure 1 Univariate - No of Employees

IV.3.1.2. Univariate Analysis For Year of Establishment

IV.3.1.2.1. Descriptive statistics for yr_of_estab:

count	25480.000
mean	1979.410
std	42.367
min	1800.000
25%	1976.000
50%	1997.000
75%	2005.000
max	2016.000

Name: yr_of_estab, dtype: float64

IV.3.1.2.2. Visual Representation for yr_of_estab:

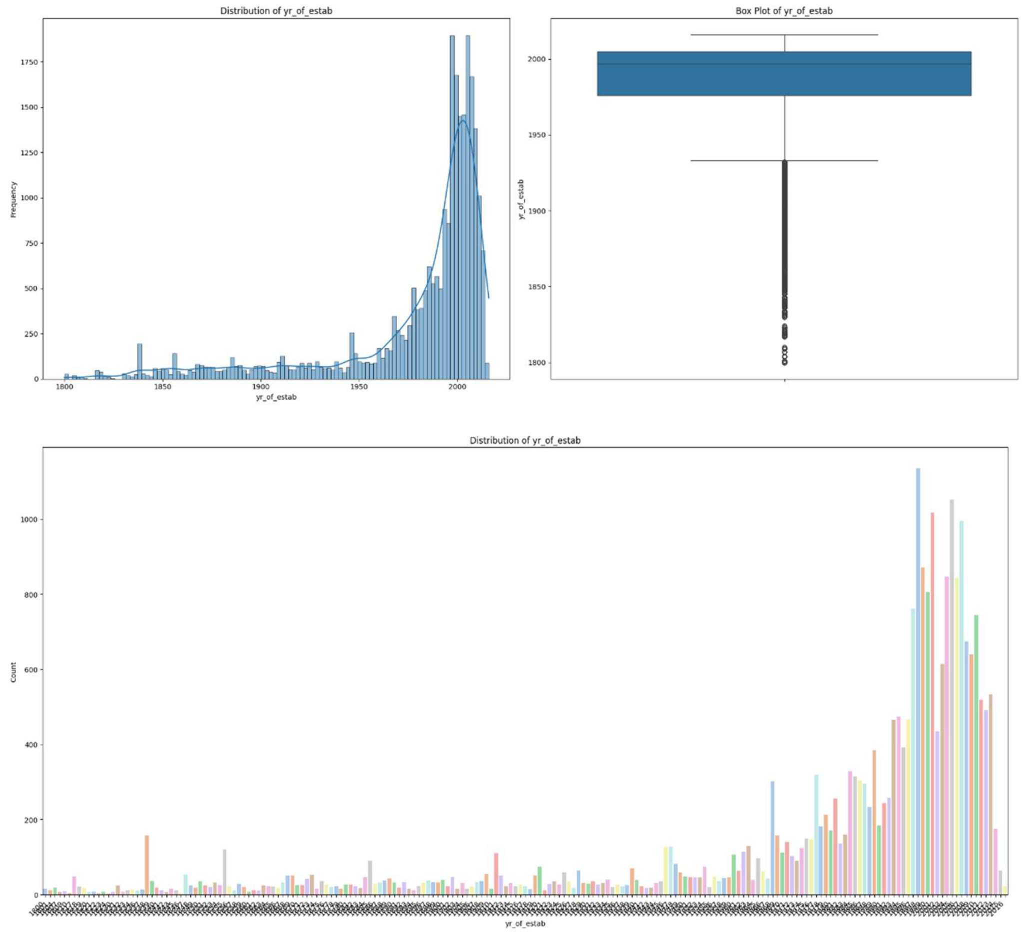


Figure 2 Univariate - Year of Establishment

IV.3.1.3. Univariate Analysis For Prevailing Wage

IV.3.1.3.1. Descriptive statistics for prevailing_wage:

count	25480.000
mean	74455.815
std	52815.942
min	2.137
25%	34015.480
50%	70308.210
75%	107735.513
max	319210.270

Name: prevailing_wage, dtype: float64

IV.3.1.3.2. Visual Representation for prevailing_wage:

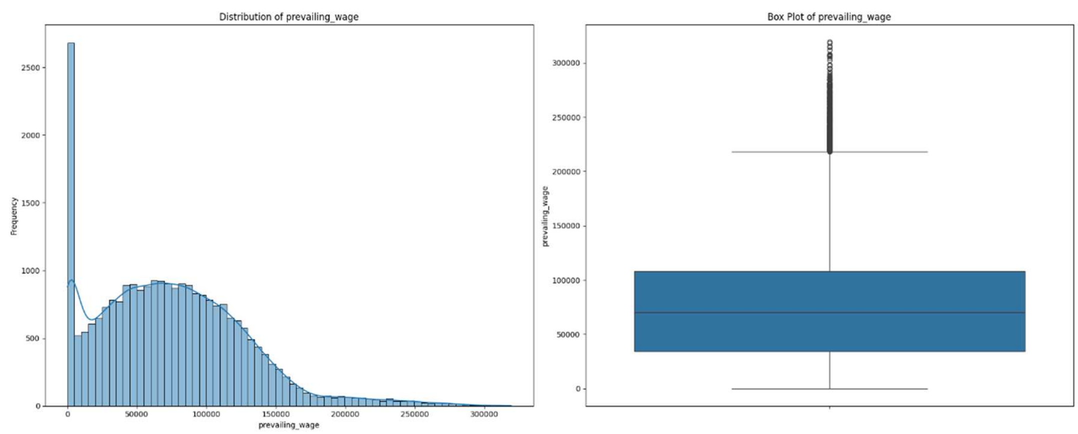
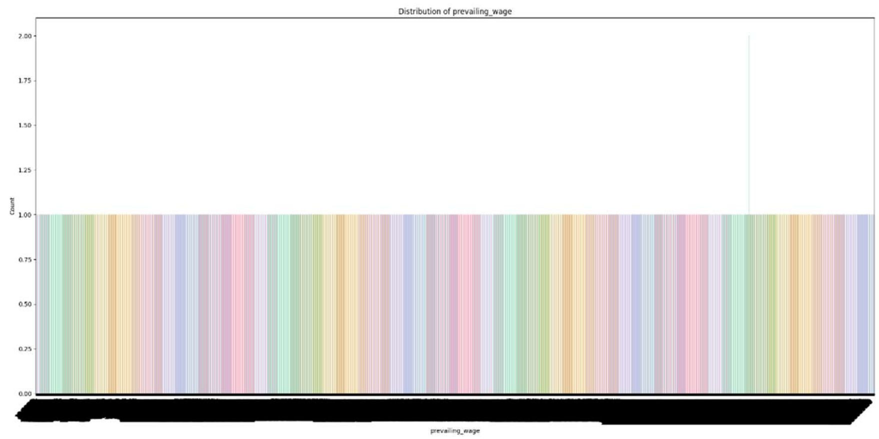


Figure 3 Univariate No of Prevailing Wage



IV.3.1.4. Inferences from Univariate Analysis for Numerical Columns

- **no_of_employees**: The number of employees varies greatly, ranging from -26 to 602,069, with a mean of approximately 5,667. The negative minimum value suggests a potential data entry error or a specific encoding that needs to be investigated; the large maximum value indicates the presence of outliers or very large companies in the dataset.
- **yr_of_estab**: The year of establishment for employers ranges from 1800 to 2016, with a mean of around 1979. This indicates a mix of very old and relatively new companies applying for visas.
- **prevailing_wage**: The prevailing wage has a wide range, from a minimum of 2.137 to a maximum of 319,210.270, with a mean of approximately 74,455. The large standard deviation (52,815) suggests a broad distribution of wages, likely influenced by different professions and regions.

IV.3.2. Univariate Analysis For Categorical columns

IV.3.2.1. Univariate Analysis For Continent

Value Counts for continent:
continent

Asia 16861
Europe 3732
North America 3292
South America 852
Africa 551
Oceania 192

Name: count, dtype: int64

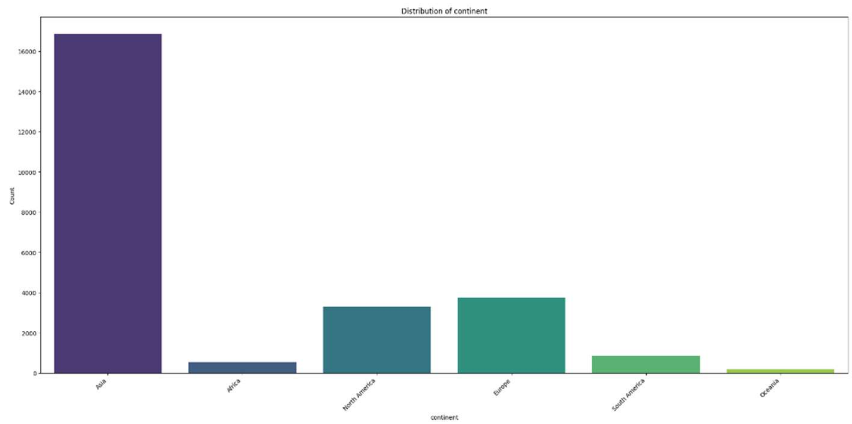


Figure 4 Univariate - Continent

IV.3.2.2. Univariate Analysis For Education of Employee

Value Counts for education_of_employee:
education_of_employee

Bachelor's	10234
Master's	9634
High School	3420
Doctorate	2192

Name: count, dtype: int64

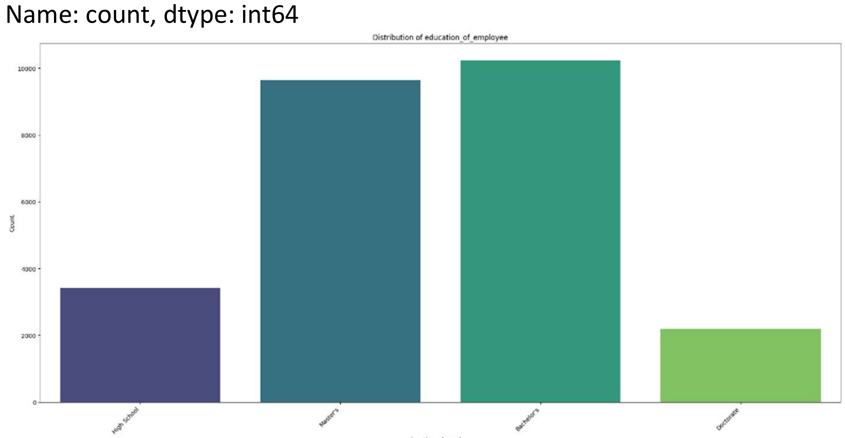


Figure 5 Univariate - Education of Employee

IV.3.2.3. Univariate Analysis For Has Job Experience

Value Counts for has_job_experience:
has_job_experience

Y	14802
N	10678

Name: count, dtype: int64

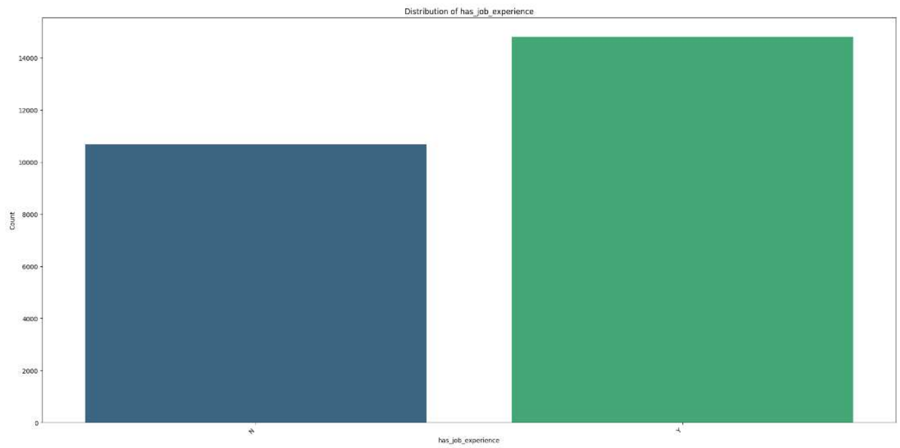


Figure 6 Univariate - Has Job Experience

IV.3.2.4. Univariate Analysis For Requires Job Training

Value Counts for requires_job_training:
requires_job_training

N	22525
Y	2955

Name: count, dtype: int64

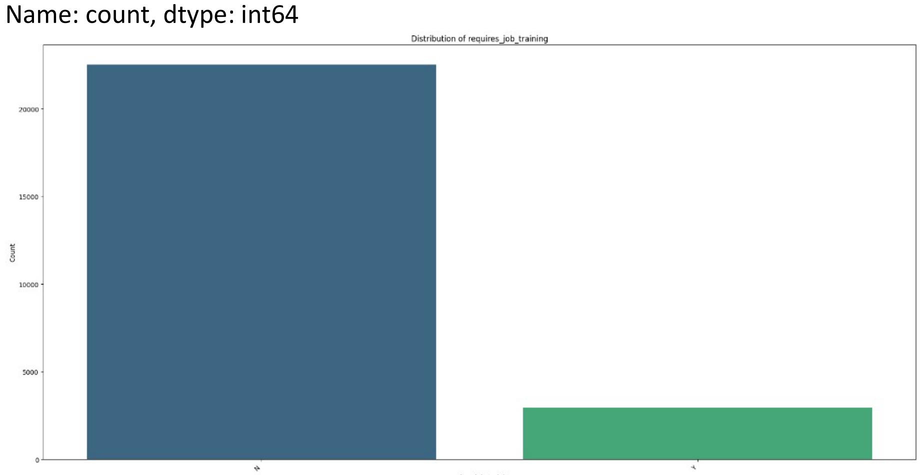


Figure 7 Univariate - Requires Job Training

IV.3.2.5. Univariate Analysis For Region of Employment

Value Counts for region_of_employment:
region_of_employment

Northeast	7195
South	7017
West	6586
Midwest	4307
Island	375

Name: count, dtype: int64

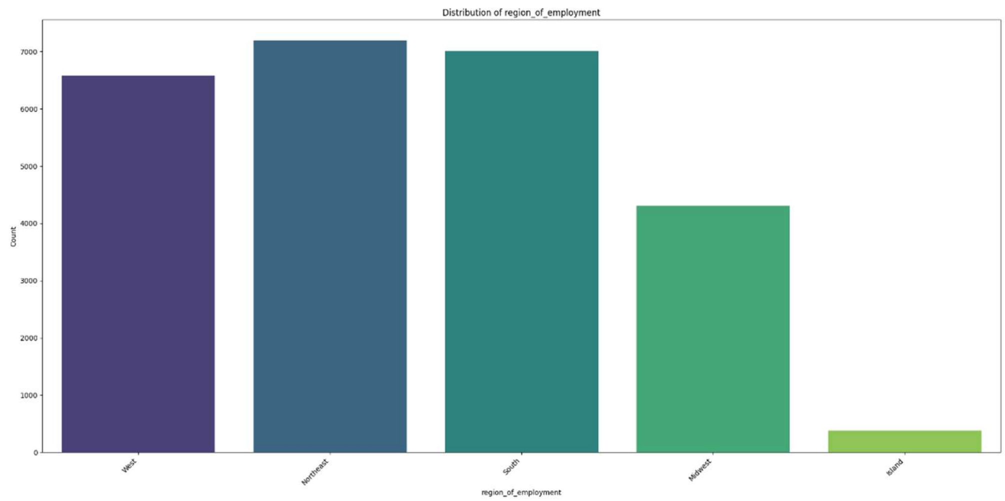


Figure 8 Univariate -Region of Employment

IV.3.2.6. Univariate Analysis For Unit of Wage

Value Counts for unit_of_wage:
unit_of_wage

Year	22962
Hour	2157
Week	272
Month	89

Name: count, dtype: int64

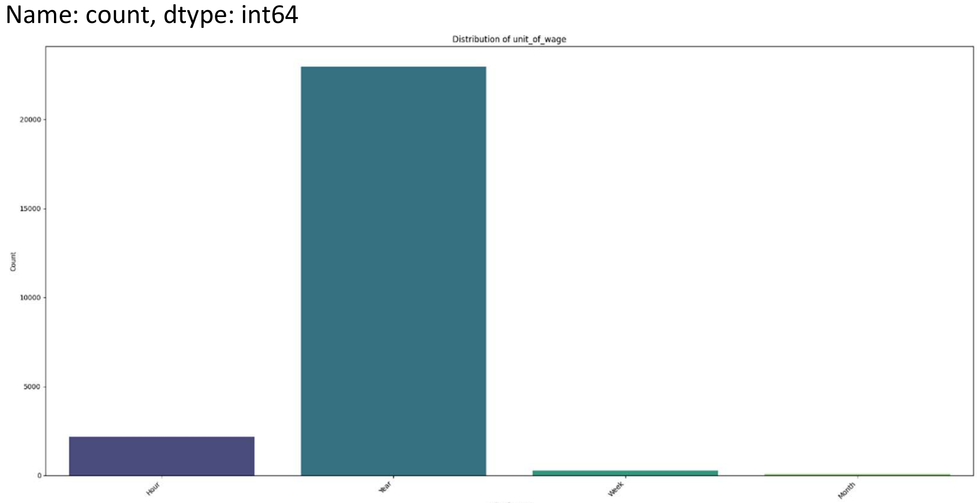


Figure 9 Univariate - Unit of Wage

IV.3.2.7. Univariate Analysis For Full Time Position

Value Counts for full_time_position:
full_time_position

Y	22773
N	2707

Name: count, dtype: int64

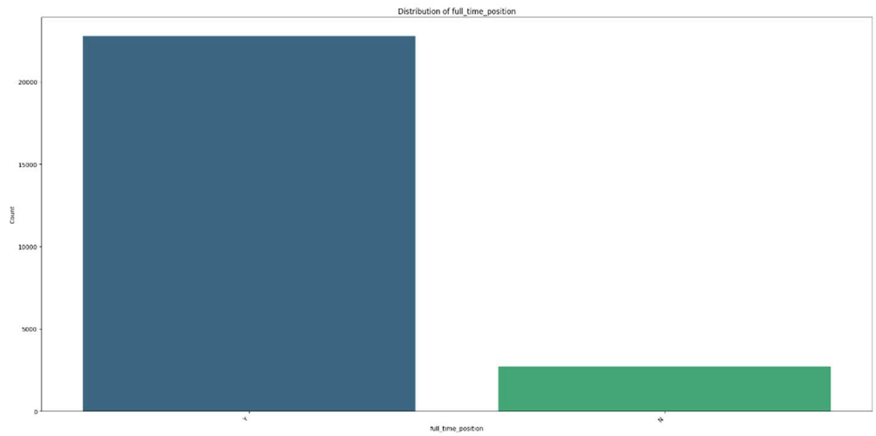


Figure 10 Univariate - Full Time Position

IV.3.2.8. Univariate Analysis For Case Status

Value Counts for case_status:
case_status

Certified	17018
Denied	8462

Name: count, dtype: int64

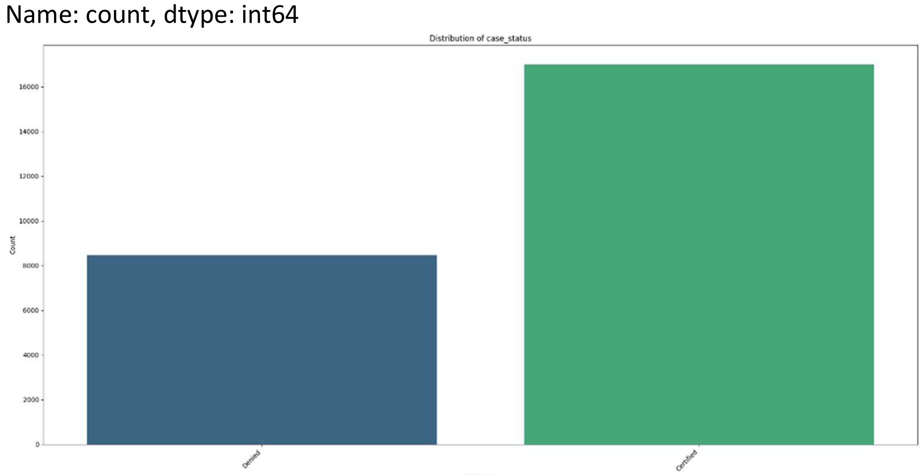


Figure 11 Univariate - Case Status

I.1.1.1. Inferences from Univariate Analysis For Categorical Columns

1. Unique values for 'continent': ['Asia' 'Africa' 'North America' 'Europe' 'South America' 'Oceania']
2. Unique values for 'education_of_employee': ['High School' "Master's" "Bachelor's" 'Doctorate']
3. Unique values for 'region_of_employment': ['West' 'Northeast' 'South' 'Midwest' 'Island']
4. Unique values for 'unit_of_wage': ['Hour' 'Year' 'Week' 'Month']
5. Unique values for 'case_status': ['Denied' 'Certified']

Observations on Percentage Distribution of Categorical Columns:

- **continent**: Asia is the dominant continent, accounting for approximately **66.17%** of applications, followed by Europe (14.65%) and North America (12.92%). Africa, South America, and Oceania have significantly smaller proportions.
- **education_of_employee**: Bachelor's and Master's degrees are the most common education levels among applicants, representing **40.16%** and **37.81%** respectively. High School and Doctorate degrees make up smaller portions.
- **region_of_employment**: The Northeast (28.24%), South (27.54%), and West (25.85%) regions of the US are the most frequently chosen for employment. The Midwest (16.90%) and Island (1.47%) regions are less common.
- **unit_of_wage**: The vast majority of wages are reported on a Yearly basis (**90.12%**), indicating that most positions are salaried. Hourly wages account for 8.47%, with Weekly and Monthly being very rare.
- **case_status**: Approximately **66.79%** of applications are Certified, while **33.21%** are Denied. This indicates a higher rate of approval than denial.
- **has_job_experience**: The majority of applicants (**58.09%**) possess job experience, indicating a preference for experienced professionals. Approximately **41.91%** do not have prior job experience.
- **requires_job_training**: A significant majority of applicants (**88.40%**) do not require job training, suggesting that employers are primarily looking for individuals ready to work immediately. Only **11.60%** require training.
- **full_time_position**: Most visa applications are for full-time positions, accounting for **89.38%**, reflecting a strong preference for permanent and dedicated roles. Part-time positions make up **10.62%** of applications.

I.2. BiVariate Analysis

I.2.1. Correlation Heatmap of Numerical Columns

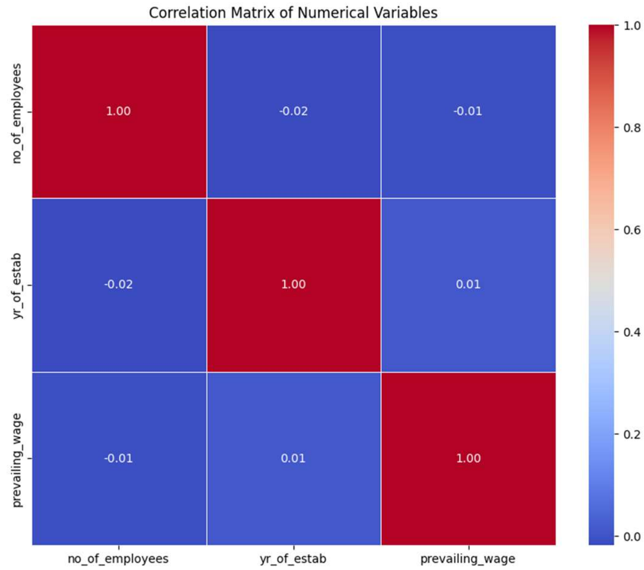


Figure 12 Correlation Map- Bivariate

	no_of_employees	**yr_of_estab**	**prevailing_wage**
no_of_employees	1.000	-0.018	-0.010
yr_of_estab	-0.018	1.000	0.012
prevailing_wage	-0.010	0.012	1.000

I.2.2. Bivariate Analysis – Numerical Vs Categorical

I.2.2.1. Case Status – Distribution

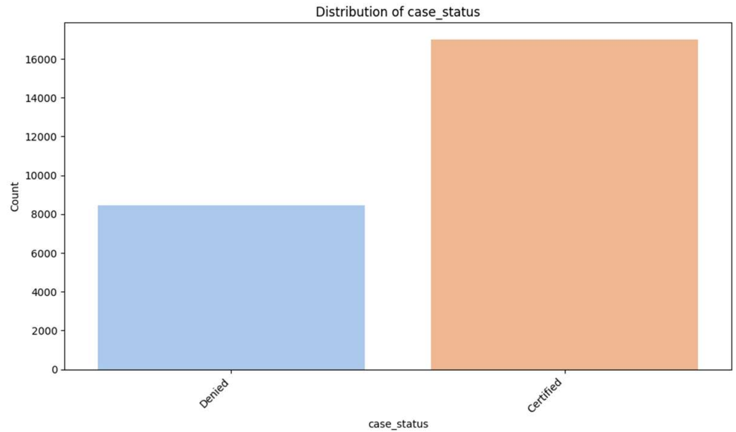


Figure 13 Case Status Distribution

I.2.2.2. No of Employees Versus Case Status

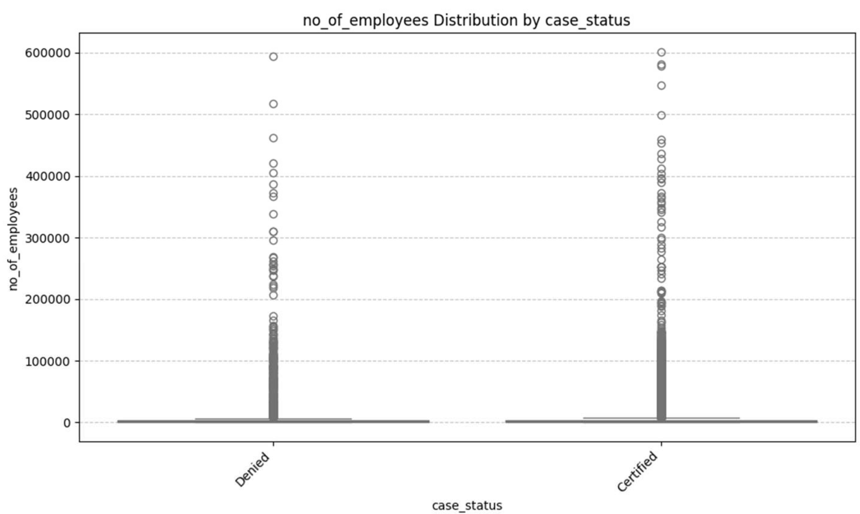


Figure 14 Bivariate - No of Employees Versus Case Status

I.2.2.3. Year of Establishment Versus Case Status

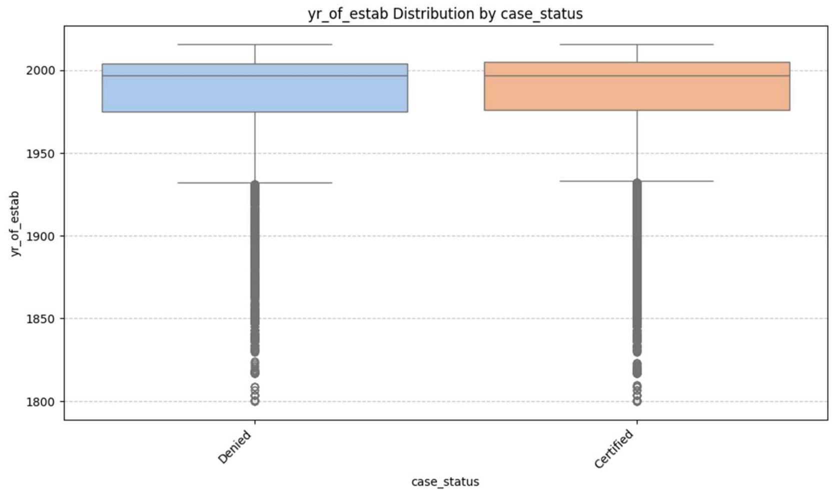


Figure 15 Bivariate -Year of Establishment Versus Case Status

I.2.2.4. Prevailing Wage Versus Case Status

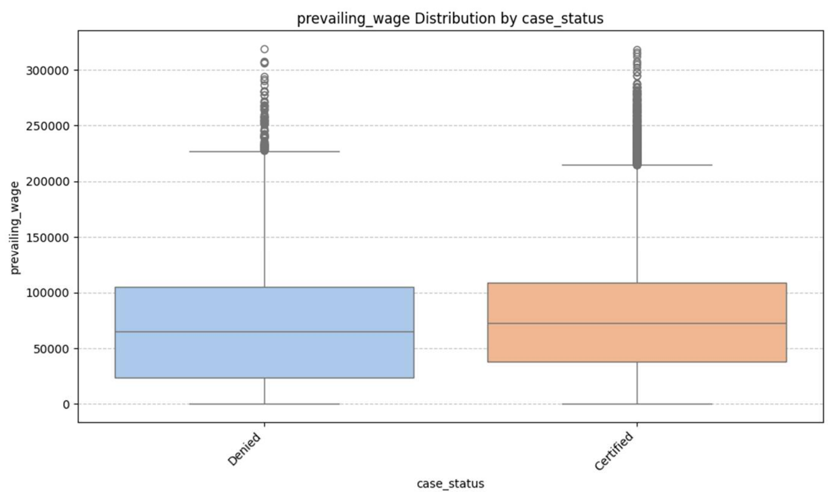


Figure 16 Bivariate - Prevailing Wage Versus Case Status

I.2.3. Inferences from Bivariate Analysis

Inferences based on the correlation matrix:

- no_of_employees and yr_of_estab: There is a very weak negative linear correlation (-0.018) between the number of employees and the year of establishment, suggesting that these two variables are largely independent of each other.
- no_of_employees and prevailing_wage: A very weak negative linear correlation (-0.010) indicates that the number of employees in a company has almost no linear relationship with the prevailing wage.

- yr_of_estab and prevailing_wage: There is a negligible positive linear correlation (0.012) between the year of establishment and the prevailing wage, implying no significant linear association.

<u>(Numerical vs. Case Status)</u>

Based on the box plots comparing numerical features with case_status:

- **no_of_employees vs. case_status:** The distribution of no_of_employees shows similar median values for both 'Certified' and 'Denied' cases. However, applications from companies with a higher number of employees tend to have a greater likelihood of being 'Certified', as indicated by the wider spread and higher maximum values in the 'Certified' category.
- **yr_of_estab vs. case_status:** There appears to be very little difference in the yr_of_estab distributions between 'Certified' and 'Denied' cases. The box plots for both categories are largely overlapping, suggesting that the year a company was established does not significantly influence the visa case_status.
- **prevailing_wage vs. case_status:** The prevailing_wage distributions differ considerably between the two case_status categories. 'Certified' applications generally show a higher median prevailing wage compared to 'Denied' applications. This suggests that applications with higher proposed wages are more frequently certified.

I.2.4. Bivariate Analysis of Categorical Vs Target Column

Categorical columns:

['continent', 'education_of_employee', 'has_job_experience', 'requires_job_training', 'region_of_employment', 'unit_of_wage', 'full_time_position', 'case_status']

Versus

case_status

I.2.4.1. Cross-tabulation for case_status by continent:

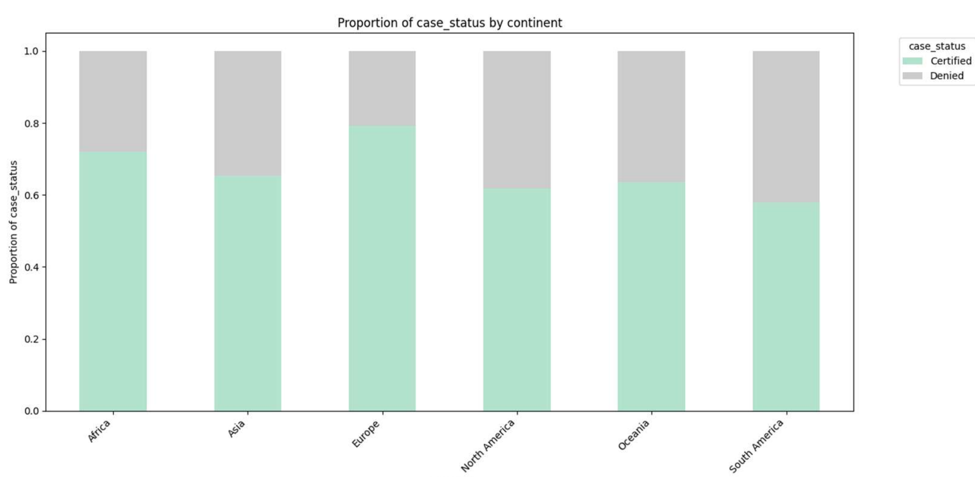


Figure 17 Bivariate - case_status by continent

Cross-tabulation for case_status by continent:

case_status	Certified	Denied
continent		
Africa	397	154
Asia	11012	5849
Europe	2957	775
North America	2037	1255
Oceania	122	70
South America	493	359

I.2.4.2. Cross-tabulation for case_status by education of employee:

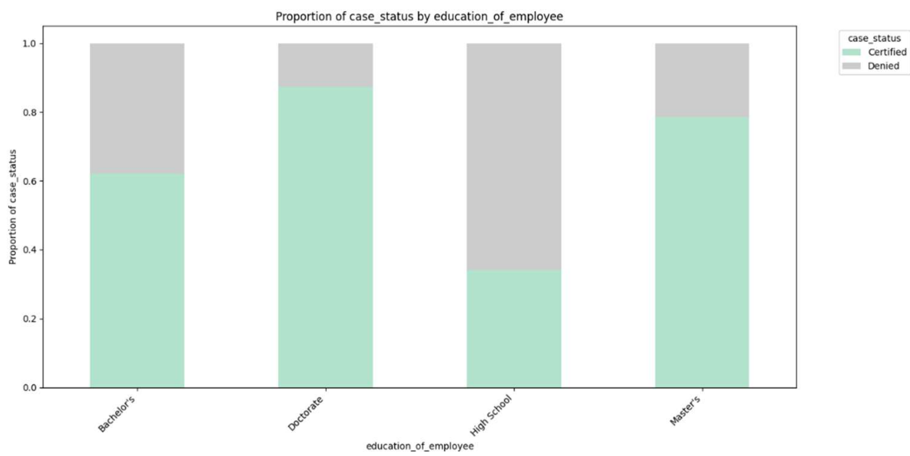


Figure 18 Bivariate - case_status by education_of_employee

Cross-tabulation for case_status by education_of_employee:

case_status	Certified	Denied
education_of_employee		
Bachelor's	6367	3867
Doctorate	1912	280
High School	1164	2256
Master's	7575	2059

I.2.4.3. Cross-tabulation for case_status by has_job_experience:

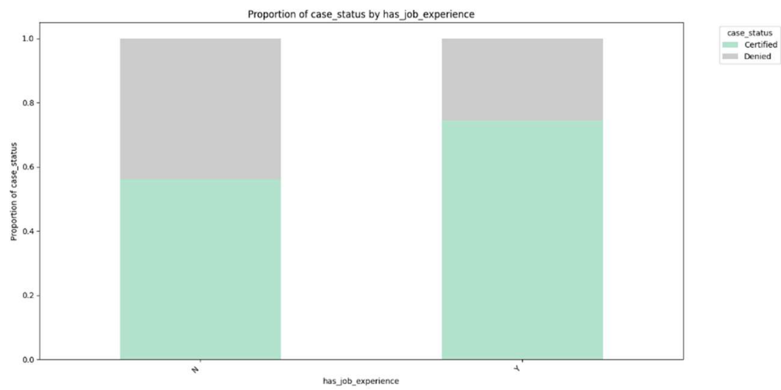


Figure 19 Bivariate - case_status by has_job_experience

Cross-tabulation for case_status by has_job_experience:

case_status	Certified	Denied
has_job_experience		
N	5994	4684
Y	11024	3778

I.2.4.4. Cross-tabulation for case_status by requires_job_training:

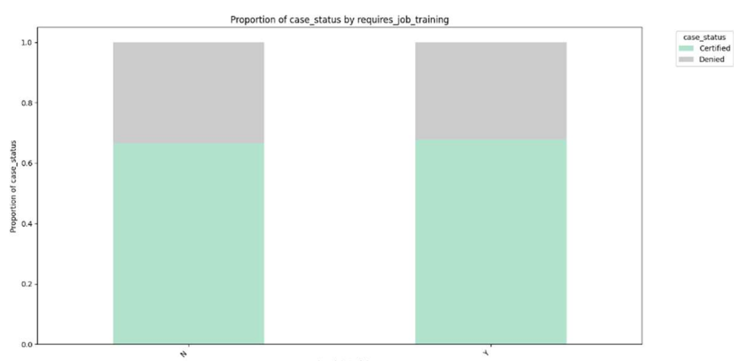


Figure 20 Bivariate - case_status by requires_job_training

Cross-tabulation for case_status by requires_job_training:

case_status	Certified	Denied
requires_job_training		
N	15012	7513
Y	2006	949

I.2.4.5. Cross-tabulation for case_status by region_of_employment:

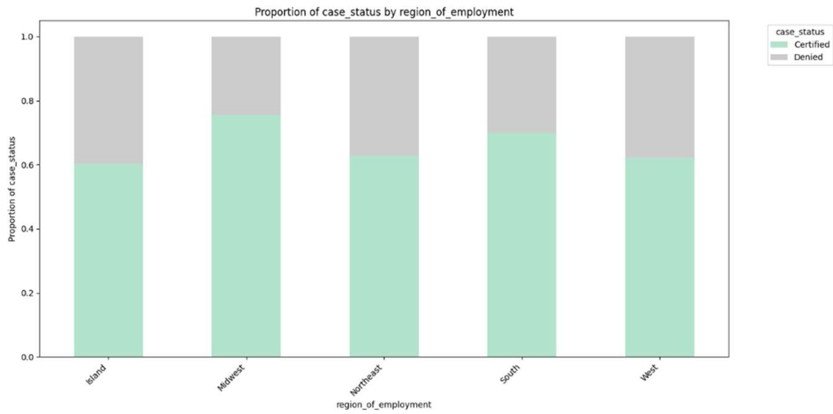


Figure 21 Bivariate - case_status by region_of_employment

Cross-tabulation for case_status by region_of_employment:

case_status	Certified	Denied
region_of_employment		
Island	226	149
Midwest	3253	1054
Northeast	4526	2669
South	4913	2104
West	4100	2486

I.2.4.6. Cross-tabulation for case_status by unit_of_wage:

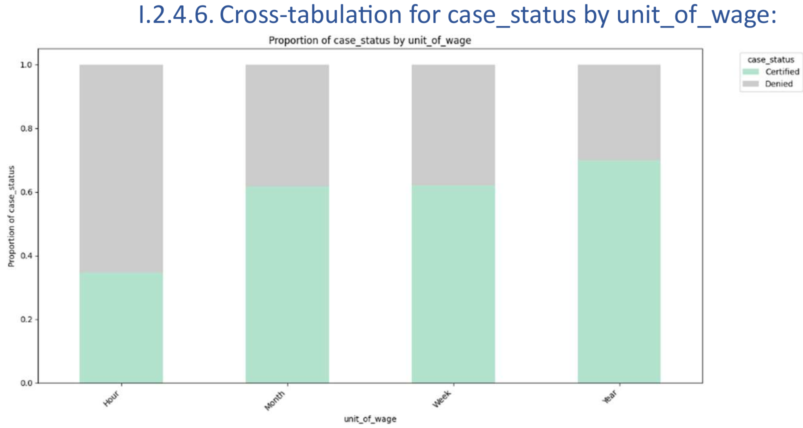


Figure 22 Bivariate - case_status by unit_of_wage

Cross-tabulation for case_status by unit_of_wage:

case_status	Certified	Denied
unit_of_wage		
Hour	747	1410
Month	55	34
Week	169	103
Year	16047	6915

I.2.4.7. Cross-tabulation for case_status by full_time_position:

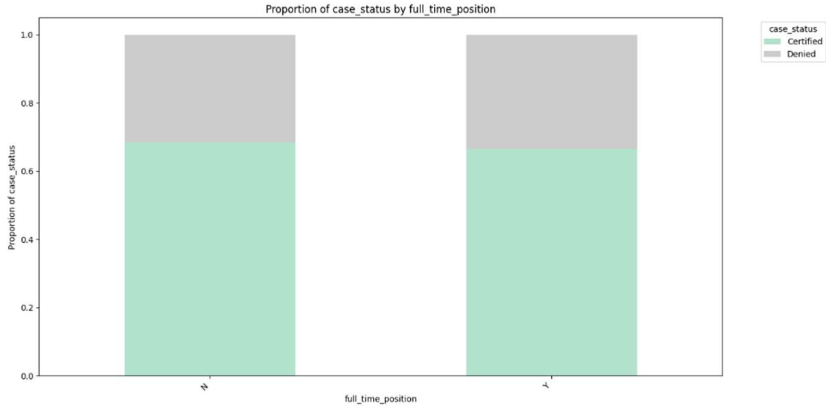


Figure 23 Bivariate - case_status by full_time_position

Cross-tabulation for case_status by full_time_position:

case_status	Certified	Denied
full_time_position		
N	1855	852
Y	15163	7610

I.2.5. Inferences from Bivariate Analysis (Categorical vs. Case Status)

Based on the cross-tabulations and stacked bar plots comparing categorical features with case_status:

- **continent vs. case_status:**
  - Applicants from **Asia** (66.17% of total applications) represent the largest group overall, but their proportion of 'Certified' cases appears to be slightly lower than other continents, or the proportion of 'Denied' cases is noticeably higher. This suggests that while volume is high, the success rate for Asian applicants might be more challenging.
  - **Europe** (14.65%) and **North America** (12.92%) might show a relatively higher proportion of 'Certified' cases, indicating a potentially smoother approval process or different applicant profiles.
  - **Africa** (2.16%), **South America** (3.34%), and **Oceania** (0.75%) might have varied proportions, but their smaller volumes make definitive statements harder without detailed percentages.
- **education_of_employee vs. case_status:**
  - Applicants with **Master's** (37.81%) and **Doctorate** (8.60%) degrees are likely to have a higher proportion of 'Certified' cases, suggesting that higher education correlates positively with visa approval.

    - **Bachelor's** degree holders (40.16%) form a large segment, and their approval rates could be slightly lower than advanced degrees but higher than 'High School'.
    - **High School** educated employees (13.42%) are expected to have the lowest proportion of 'Certified' cases, indicating that a higher level of education is generally preferred for visa certification.
- **has_job_experience vs. case_status:**
    - Employees who **have job experience (Y - 58.09% of applicants)** are highly likely to have a higher proportion of 'Certified' visas compared to those who **do not have job experience (N - 41.91% of applicants)**. This indicates that prior work experience is a significant factor in visa approval.
- **requires_job_training vs. case_status:**
    - Applicants who **do not require job training (N - 88.40% of applicants)** are expected to have a significantly higher proportion of 'Certified' visas than those who **require job training (Y - 11.60% of applicants)**. This suggests employers and the visa process prefer candidates who are immediately ready for work.
- **region_of_employment vs. case_status:**
    - Certain regions like **Northeast** (28.24%), **South** (27.54%), and **West** (25.85%) might show varying proportions of 'Certified' vs. 'Denied' cases, possibly reflecting regional labor demands or economic conditions.
    - There could be subtle differences in approval rates across regions, indicating that the location of intended employment might play a role in visa outcomes.
- **unit_of_wage vs. case_status:**
    - Applications with a **Yearly** unit of wage (90.12% of applications) might have a higher proportion of 'Certified' cases, as these often correspond to more stable, higher-paying, and full-time professional positions.
    - **Hourly** (8.47%), **Weekly** (1.07%), and **Monthly** (0.35%) wage units could show different approval rates, potentially lower for hourly wages, reflecting different types of employment or perceived job stability.
- **full_time_position vs. case_status:**
    - Applications for **full-time positions (Y - 89.38% of applications)** are very likely to have a much higher proportion of 'Certified' cases than those for **part-time positions (N - 10.62% of applications)**. This strongly suggests a preference for full-time employment in the visa approval process.

Overall, these bivariate analyses highlight several key drivers of visa case_status, with has_job_experience, requires_job_training, education_of_employee, and full_time_position appearing to be particularly influential in determining whether a visa is 'Certified' or 'Denied'.

I.3. Key meaningful observations on individual variables and the relationship between variables

Based on the comprehensive univariate and bivariate analyses, several critical observations emerge regarding the characteristics of visa applicants and employers, and their influence on visa case_status. These insights can guide EasyVisa in developing data-driven recommendations.

Observations on Individual Variables (Univariate Analysis):

- **case_status**: The overall visa approval rate is approximately **66.79% Certified** versus **33.21% Denied**. This indicates that while a majority are approved, a significant portion faces denial, highlighting the need for predictive modeling.
- **continent**: **Asia** is the dominant source of applications, accounting for approximately **66.17%** of all applicants. Europe (14.65%) and North America (12.92%) follow, suggesting major talent flows from these regions.
- **education_of_employee**: A strong preference for educated talent is evident, with **Bachelor's (40.16%)** and **Master's (37.81%)** degree holders forming the largest groups. High school graduates are least represented (13.42%).
- **has_job_experience**: The majority of applicants (**58.09%**) possess prior job experience, aligning with employers' likely preference for skilled individuals.
- **requires_job_training**: Most applicants (**88.40%**) do not require job training, implying a demand for candidates ready to contribute immediately.
- **region_of_employment**: The **Northeast (28.24%), South (27.54%), and West (25.85%)** are the most frequent regions for employment, reflecting concentrated labor demands.
- **unit_of_wage**: Nearly all wages are reported on a **Yearly (90.12%)** basis, indicating that most positions are salaried and professional.
- **full_time_position**: An overwhelming majority of applications (**89.38%**) are for full-time positions, signifying a strong preference for permanent roles.
- **no_of_employees**: The number of employees varies significantly (from -26 to 602,069). The presence of a negative value (-26) suggests a potential data entry error or specific encoding that requires careful handling during preprocessing.
- **yr_of_estab**: Companies range from very old (1800) to relatively new (2016), with a mean around 1979.
- **prevailing_wage**: Wages exhibit a broad distribution (from 2.137 to 319,210.270), reflecting diverse roles and regional economic variations.

Observations on Relationships (Bivariate Analysis):

Numerical Features vs. case_status:

- **no_of_employees vs. case_status**: Applications from companies with a **higher number of employees** tend to have a greater likelihood of being Certified. This might suggest that larger, more established companies have better resources or more stringent internal processes that lead to higher approval rates.

- **yr_of_estab vs. case_status**: The year of establishment shows **little to no significant influence** on case_status. Both older and newer companies experience similar approval and denial rates.
- **prevailing_wage vs. case_status**: Applications with a **higher prevailing_wage** generally have a greater probability of being Certified. This suggests that higher-paying positions, often indicative of specialized skills or higher demand, are more likely to secure visa approval.

Categorical Features vs. case_status:

- **continent vs. case_status**: While **Asia** accounts for the largest volume of applications, their proportion of Certified cases might be slightly lower, or Denied cases noticeably higher, compared to other continents. Applicants from **Europe and North America** might experience relatively smoother approval processes. This points to potential underlying differences in application quality, occupational demand, or processing biases across continents.
- **education_of_employee vs. case_status**: Higher educational attainment is a significant driver for approval. Applicants with **Master's and Doctorate degrees** show a considerably higher proportion of Certified cases, whereas **High School** educated employees have the lowest certification rates. This confirms that advanced education is a strong positive indicator for visa approval.
- **has_job_experience vs. case_status**: Employees who **have job experience (Y)** are highly likely to have their visas Certified. Prior work experience is a very strong positive determinant in visa outcomes.
- **requires_job_training vs. case_status**: Applicants who **do not require job training (N)** have a significantly higher proportion of Certified visas. This indicates a preference for candidates who are immediately deployable without further employer investment in training.
- **region_of_employment vs. case_status**: There are subtle differences in approval rates across various regions, suggesting that local labor market demands, economic conditions, or specific regional policies might play a role in visa outcomes.
- **unit_of_wage vs. case_status**: Applications with a **Yearly** unit of wage are associated with a higher proportion of Certified cases, reinforcing the idea that stable, full-time professional positions are more favorably viewed in the visa process.
- **full_time_position vs. case_status**: Applications for **full-time positions (Y)** are overwhelmingly more likely to be Certified than part-time positions. This strongly suggests that the visa process prioritizes long-term, full-time employment commitments.

Overall Business Insights:

These analyses collectively highlight that **education level, prior job experience, not requiring job training, full-time employment, and a higher prevailing wage** are the most influential factors driving visa certification. EasyVisa should focus on these key drivers to recommend optimal applicant profiles and improve the chances of visa approval. Furthermore, understanding the nuances of continent-specific outcomes and the implications of employer size (no_of_employees) can inform more targeted strategies for applicants.

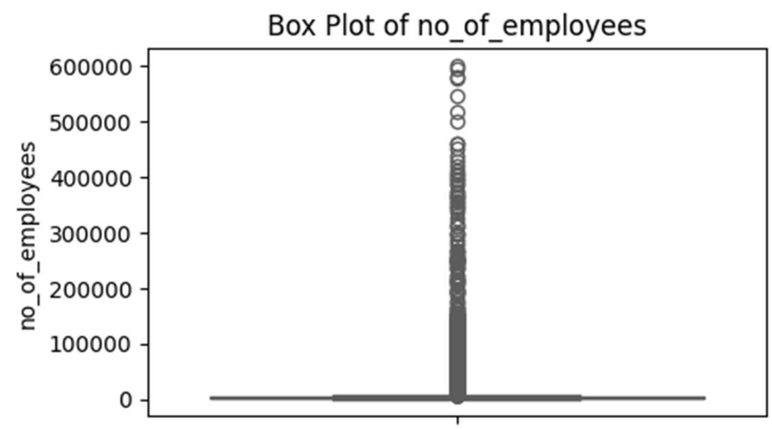
II. Data Preprocessing

II.1. Missing Value Treatment

There are no missing values in data and hence no missing value treatment is required

There are 25480 rows and 12 columns. There are no duplicate ones.

II.2. Outlier Detection and Treatment



Box Plots to check the outliers:

Figure 24 Outlier - No of Employees

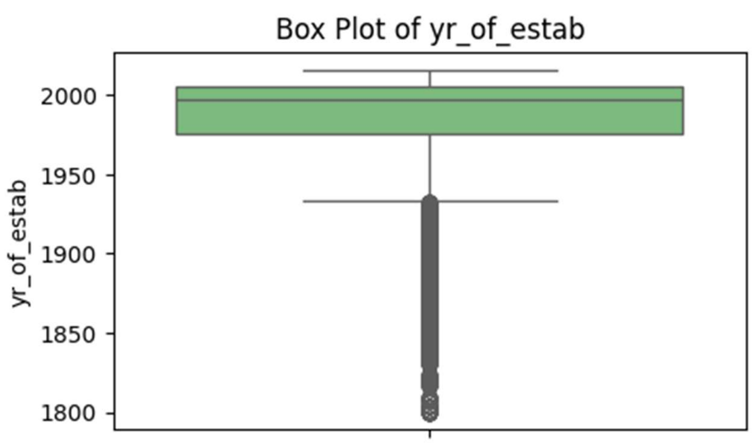


Figure 25 Outlier - Year of Establishment

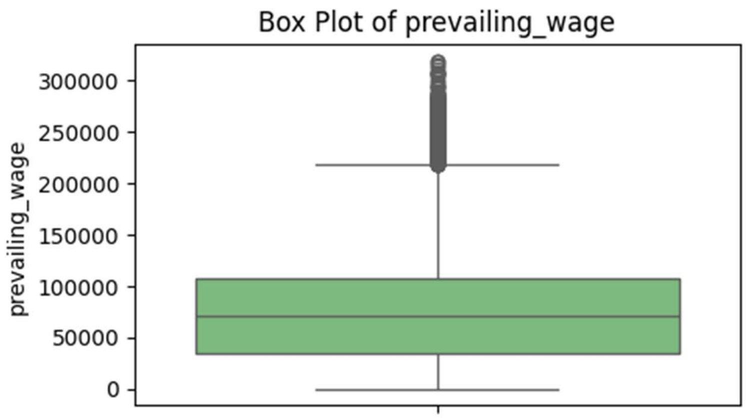


Figure 26 Outlier Check - Box Plot For Prevailing Wage

Inferences on Outliers from Box Plots

Based on the box plots generated for the numerical columns, we can observe the following regarding outliers:

- `no_of_employees`: The box plot for `no_of_employees` shows a significant number of outliers on the higher side, indicating the presence of a few very large companies. There is also an unusual negative minimum value (-26), which is likely a data entry error or an artifact of data collection, suggesting the need for data cleaning or specific handling during modeling. This is handled in next steps in this during feature engineering step.
- `yr_of_estab`: The `yr_of_estab` box plot reveals outliers on the lower side, corresponding to companies established in the very early years (e.g., 1800s). This indicates a mix of very old, established companies alongside more recent ones. These older values, while outliers, are likely genuine data points. Tenure of establishment is calculated later in feature engineering step.
- `prevailing_wage`: The `prevailing_wage` box plot shows numerous outliers on both the lower and higher ends. This suggests a wide range of wages, from very low to very high, which is typical for diverse job roles and industries. These outliers are likely reflective of genuine variations in compensation rather than errors. Also we are not sure of exact hours per day or hours per week, used in data collection, so we cannot convert to yearly for other data like hourly, weekly etc.

Business Inference & Modeling Consideration:

- **Impact on Models:** Outliers in `no_of_employees` and `prevailing_wage` can significantly influence models that are sensitive to extreme values (e.g., Linear Regression, Logistic Regression without robust scaling). Tree-based models (like Decision Trees) are generally more robust to outliers as they focus on splits based on feature values rather than distances or magnitudes.
- **Data Quality:** The negative value in `no_of_employees` definitely warrants investigation. For `yr_of_estab`, the old values are likely valid and represent the longevity of some companies.
- **Decision for Treatment:** Given that **Decision Tree based models are inherently robust to outliers**, explicit outlier capping or removal may not be strictly necessary for these models. For models like Logistic Regression, applying **`RobustScaler`** (as done in `df_reg` feature

engineering) is a suitable strategy as it scales features based on medians and interquartile ranges, making it less susceptible to the influence of extreme values. This approach avoids altering the underlying data distribution while mitigating the impact of outliers on model training.

II.3. Feature Engineering

Duplicate the data frame and then do the feature engineering

II.3.1.Feature Engineering

Feature Engineering Before the Train-Validation-Test split:

1. Drop the unique identifier, as this doesn't help us in analysis or model building. Remove the case_id
2. No of employees have negative values. This is assumed as the data entry error, as the no of employees cannot be negative. Convert negative values of no_of_employees to absolute values
3. Year of establishment will help us to find the age or tenure of company . So we can convert yr_of_estab to the tenure with calculation as (Current year(2026) - yr_of_estab)
4. Label Encoding of has_job_experience values -> 'N': 0, 'Y': 1
5. Label Encoding of requires_job_training values -> 'N': 0, 'Y': 1
6. Label Encoding of full_time_position values -> 'N': 0, 'Y': 1
7. Ordinal Encoding of education_of_employee values -> 'High School': 0, "Bachelor's" : 1, "Master's": 2, "Doctorate", 3
8. Label Encoding of Target Variable case_status values -> 'Denied': 0, 'Certified': 1
9. Conversion of data type for the remaining object types to category: data types of continent, region_of_employment, unit_of_wage should be category

The columns after Feature Engineering in this case are as below:

Data columns (total 11 columns):

Column Non-Null Count Dtype
0 continent 25480 non-null category
1 education_of_employee 25480 non-null int64
2 has_job_experience 25480 non-null int64
3 requires_job_training 25480 non-null int64
4 no_of_employees 25480 non-null int64
5 region_of_employment 25480 non-null category

6	prevailing_wage	25480 non-null	float64
7	unit_of_wage	25480 non-null	category
8	full_time_position	25480 non-null	int64
9	case_status	25480 non-null	int64
10	tenure_of_estab	25480 non-null	int64

Feature Engineering After the Train-Validation-Test split:

1. One hot Encoding for category type columns: continent, region_of_employment, unit_of_wage. This is performed after splitting to prevent data leakage from the test or validation sets into the training process during encoding.

```
One-Hot Encoding applied to X_train, X_val, and X_test.
X_train_encoded shape: (20384, 19): 20384 rows and 19 columns
X_val_encoded shape: (1274, 19): 1274 rows and 19 columns
X_test_encoded shape: (3822, 19): 3822 rows and 19 columns
```

The columns after Feature Engineering in this case are as below:

```
Data columns :

['education_of_employee', 'has_job_experience', 'requires_job_training', 'no_of_employees',
'prevailing_wage', 'full_time_position', 'case_status', 'tenure_of_estab', 'continent_Asia',
'continent_Europe', 'continent_North America', 'continent_Oceania', 'continent_South America',
'region_of_employment_Midwest', 'region_of_employment_Northeast',
'region_of_employment_South', 'region_of_employment_West', 'unit_of_wage_Month',
'unit_of_wage_Week', 'unit_of_wage_Year']
```

II.4. Data Scaling

Feature scaling is not required because these models are not sensitive to the magnitude or scale of your features.

Here's why, and a breakdown for each model:

- Decision Tree: Decision Trees work by splitting nodes based on a single feature's value. The decision is purely about whether a value is greater than or less than a threshold. The actual scale of the feature doesn't change the split points or the resulting tree structure.
- Bagging Classifier: Bagging (like BaggingClassifier) typically uses Decision Trees as its base estimators. Since the underlying Decision Trees don't require scaling, neither does the ensemble that bags them.
- Boosting Classifiers (general term including AdaBoost, Gradient Boosting, XGBoost): These algorithms also primarily rely on Decision Trees (often shallow ones, called 'weak learners'). The boosting process focuses on sequentially improving predictions by giving more weight to misclassified instances. The splitting logic of the weak learners remains insensitive to feature scales.
- AdaBoost: As a boosting algorithm using Decision Trees, AdaBoost does not require feature scaling.

- RandomForest: Random Forests are an ensemble of many Decision Trees. Each tree independently makes splits based on feature values, so scaling doesn't affect their performance.
- XGBoostClassifier: XGBoost (Extreme Gradient Boosting) is a highly optimized gradient boosting framework that also uses Decision Trees as its base learners. Therefore, feature scaling is not typically necessary.
- GradientBoost (GradientBoostingClassifier): Similar to XGBoost and AdaBoost, GradientBoostingClassifier builds an ensemble of Decision Trees. Feature scaling is not a prerequisite for this model either.

In summary, for Decision Trees, Random Forests, Bagging, and all boosting algorithms (AdaBoost, Gradient Boosting, XGBoost), we can generally proceed without applying feature scaling.

II.5. Train-Validation-test split

We have used an 80/15/5 split for training, testing, and validation respectively.

- There are **20384 rows** in the training data.
- There are **3822 rows** in the test data.
- There are **1274 rows** in the validation data.

Training dataset is used to train the model.

Validation dataset is used for hyper tuning of the model

Test dataset is used for testing the model.

III. Model Building

III.1.1. Choosing the Metric to Optimize for the Problem

The primary objective for the EasyVisa problem is to facilitate visa approvals and recommend suitable profiles based on drivers influencing case status. This is a binary classification problem where we are trying to predict if a visa application will be 'Certified' (positive class) or 'Denied' (negative class).

Let's consider the implications of different types of errors:

- **False Positives (Type I Error):** The model predicts a visa will be **Certified**, but it is actually **Denied**. This means an unsuitable candidate might be recommended, leading to wasted processing time and resources.
- **False Negatives (Type II Error):** The model predicts a visa will be **Denied**, but it actually **Is Certified**. This means a deserving candidate might be overlooked or not recommended, potentially missing out on valuable talent.

Given the objective of facilitating approvals and recommending suitable profiles, it is critical for the model to correctly identify candidates who *will be certified* while minimizing false negatives. Missing out on a certified candidate (False Negative) can be seen as a lost opportunity for both the applicant and the employers in the US. However, recommending too many unsuitable candidates (False Positives) can also overwhelm the system.

Therefore, a balanced approach is needed. The most suitable metrics to optimize for this problem are:

1. **Recall (Sensitivity) for the 'Certified' class:** This measures the proportion of actual certified visas that were correctly identified by the model. A high recall means the model is effective at catching most of the applications that will actually be certified, thereby allowing the system to identify deserving candidates.
2. **F1-Score for the 'Certified' class:** This is the harmonic mean of Precision and Recall. It provides a balance between minimizing False Negatives (Recall) and minimizing False Positives (Precision). While Recall is important to find most certified cases, a reasonable Precision is also important to avoid recommending too many ultimately denied cases, which would make the recommendation system inefficient.

Inference: While **Recall** is extremely important to catch as many certified applications as possible, aiming for a high **F1-Score** for the 'Certified' class provides a balanced view. It ensures that while we are catching most actual certifications, we are not doing so at the expense of an unacceptably high number of false recommendations. We will primarily focus on maximizing the F1-Score, while also closely monitoring Recall for the positive class. The **Confusion Matrix** will also be a critical tool for detailed analysis of the model's performance in identifying 'Certified' applications.

ROC AUC is valuable for understanding the overall discriminatory power of model, and a high AUC is a strong indicator of a good model. We will check this as well.

III.1.2. Choose Model and Approach:

Our approach to solve this business problem, is to choose the various models given below and find the metrics.

1. Decision Tree
2. Bagging Classifier
3. AdaBoostClassifier
4. GradientBoost Classifier
5. XGBClassifier
6. RandomForestClassifier

We will try with various datasets. We will use training dataset to train model, validate dataset to hyper tune models, test dataset finalise to run the chosen best few models.

1. Original dataset
2. UnderSampled data
3. OverSampled data

We will do the hypertuning on the few good models selected.

There are 18 combinations (6 models on Original dataset, 6 models on UnderSampled data and 6 models on OverSampled data). In ideal world, we can pick up the best 3-4 models and then do hypertuning. Here, for completeness, the values from all of these are checked and provided. Then we check the initial models and hypertuned models for the best metrics and then decide the best models and recommend.

III.1.3. Model Building

III.1.3.1. Model Building -Original DataSet

III.1.3.1.1. Cross-Validation Scores (Certified Class):

The output for the K-fold (5 splits used here) , with scoring metrics (f1 score and recall) is as follows:

- Decision Tree:
  - F1-Score: 0.7440
  - Recall: 0.7415
- Bagging:
  - F1-Score: 0.7745
  - Recall: 0.7746
- AdaBoost:
  - F1-Score: 0.8190
  - Recall: 0.8859
- GradientBoost:
  - F1-Score: 0.8245
  - Recall: 0.8744
- XGBoost:
  - F1-Score: 0.8101
  - Recall: 0.8547
- RandomForest:
  - F1-Score: 0.8025
  - Recall: 0.8387

III.1.3.1.2. Training and Validation Performance:

1. Decision Tree

Training Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	1.000	1.000	1.000	1.000	1.000

Training Set Confusion Matrix:

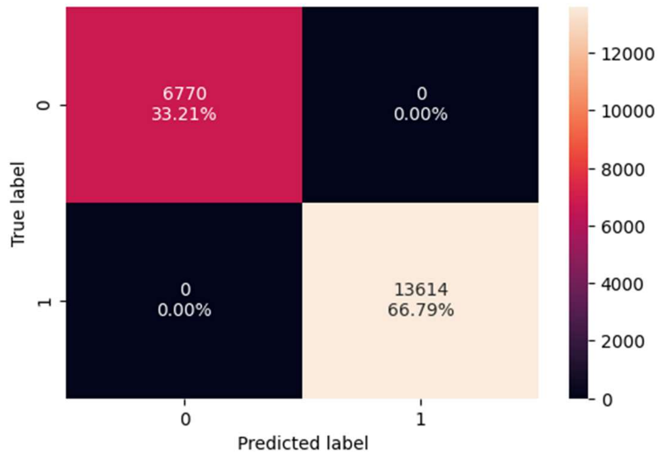


Figure 27 Confusion Matrix- Original - DecisionTree

Validation Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.649	0.727	0.742	0.735	0.610

Validation Set Confusion Matrix:

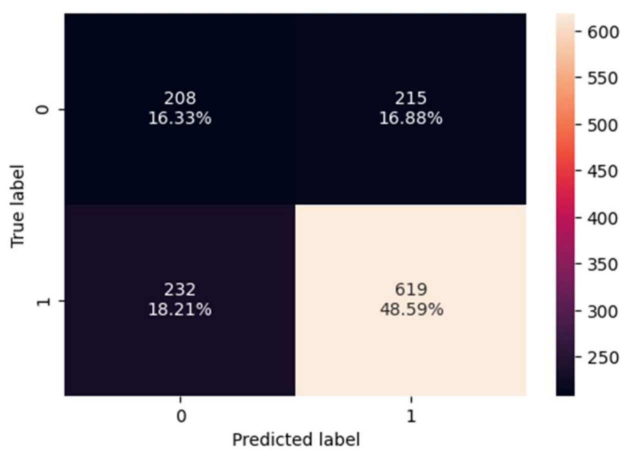


Figure 28 Confusion Matrix- Original - Validation -DecisionTree

2. Bagging

Training Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.984	0.985	0.991	0.988	0.998

Training Set Confusion Matrix:

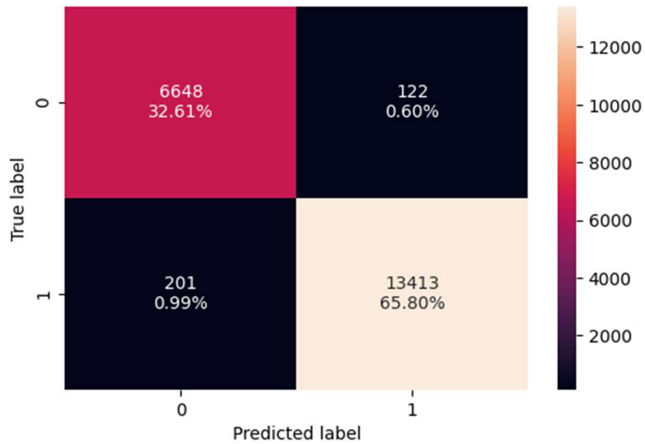


Figure 29 Confusion Matrix- Original - Training - Bagging

Validation Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.683	0.759	0.764	0.762	0.708

Validation Set Confusion Matrix:

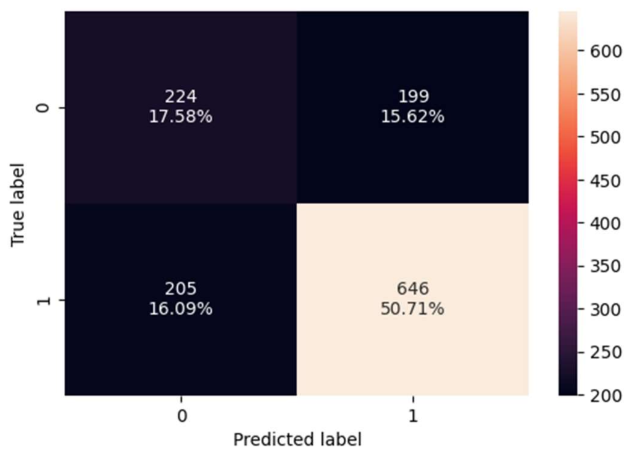


Figure 30 Confusion Matrix- Original - Validation - Bagging

3. AdaBoost

Training Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.739	0.885	0.762	0.819	0.774

Training Set Confusion Matrix:

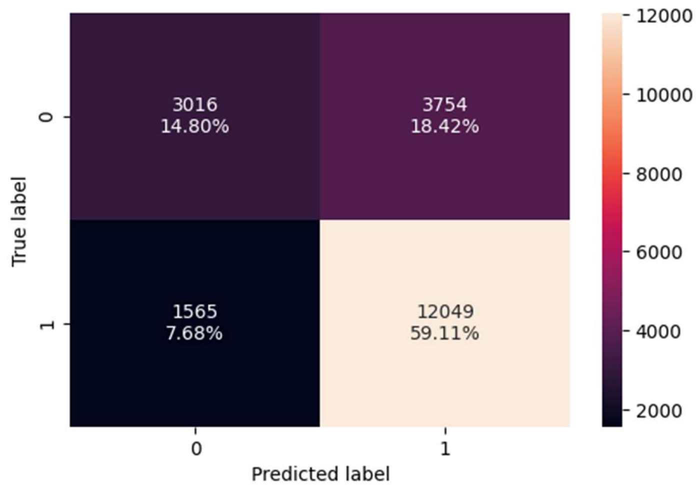


Figure 31 Confusion Matrix- Original - Training - AdaBoost

Validation Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.732	0.886	0.756	0.816	0.743

Validation Set Confusion Matrix:

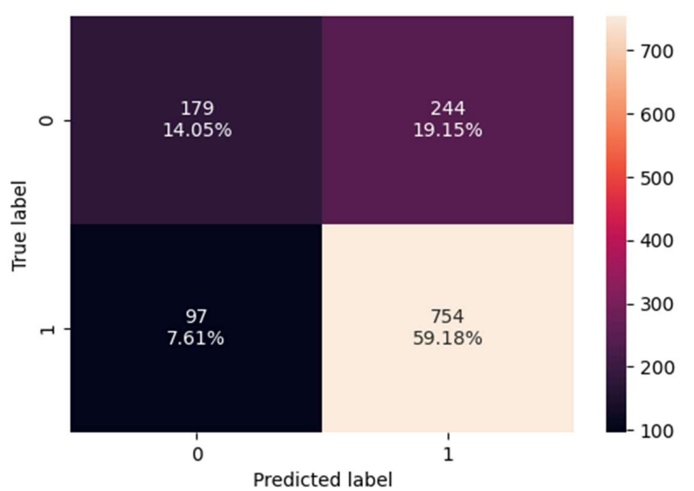


Figure 32 Confusion Matrix- Original - Validation - AdaBoost

4. GradientBoost

Training Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.757	0.878	0.784	0.829	0.795

Training Set Confusion Matrix:

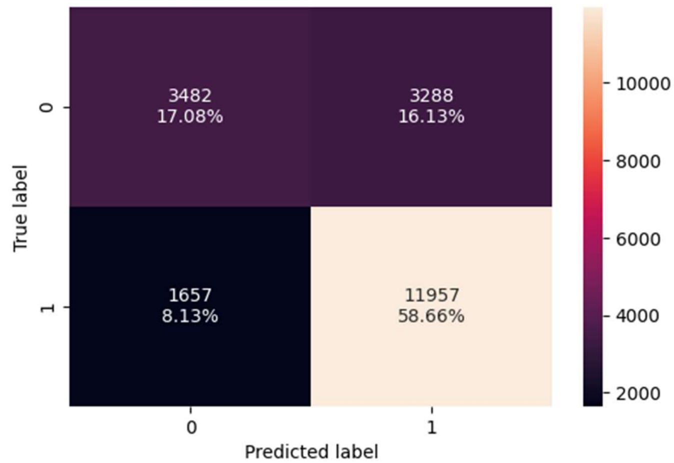


Figure 33 Confusion Matrix- Original - Training - GradientBoost

Validation Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.743	0.879	0.769	0.820	0.752

Validation Set Confusion Matrix:

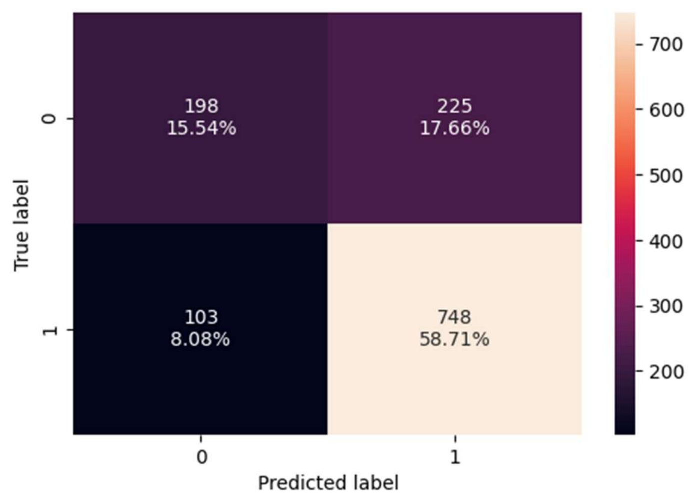


Figure 34 Confusion Matrix- Original - Validation - GradientBoost

5. XGBoost

Training Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.834	0.926	0.841	0.882	0.908

Training Set Confusion Matrix:

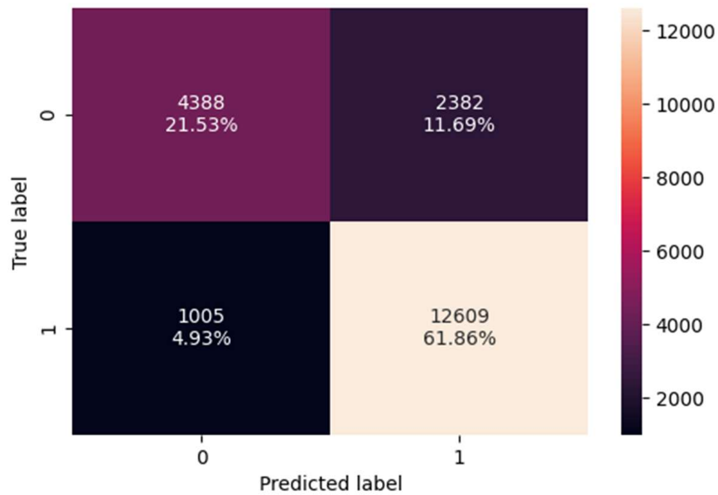


Figure 35 Confusion Matrix- Original - Training - XGBoost

Validation Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.732	0.868	0.763	0.812	0.748

Validation Set Confusion Matrix:

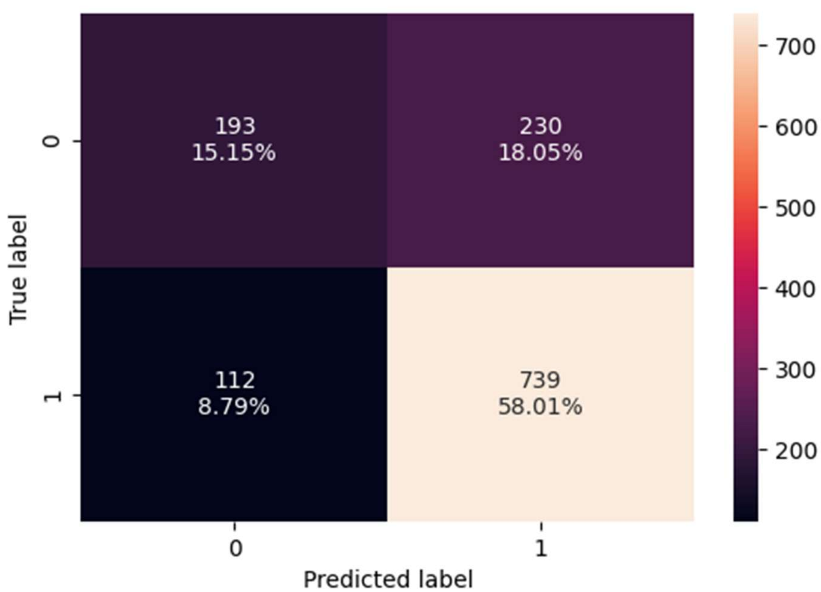


Figure 36 Confusion Matrix- Original - Validation - XGBoost

6. RandomForest

Training Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	1.000	1.000	1.000	1.000	1.000

Training Set Confusion Matrix:

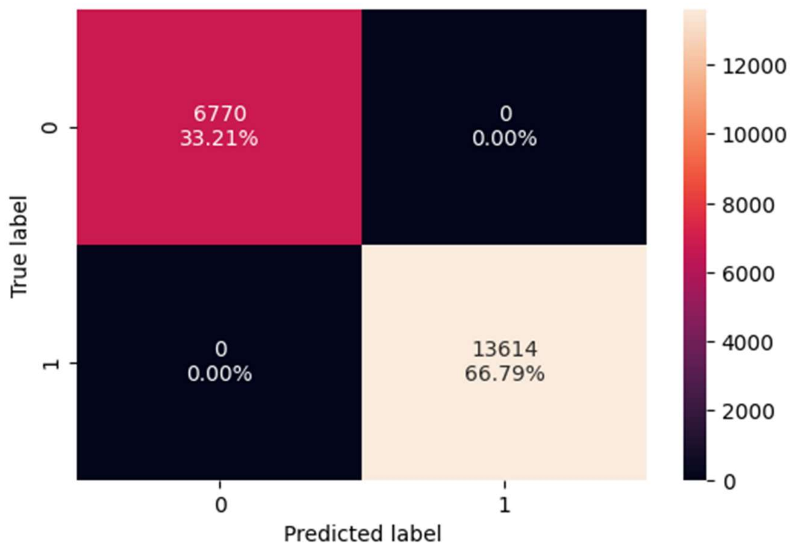


Figure 37 Confusion Matrix- Original - Training - RandomForest

Validation Set Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.716	0.844	0.758	0.799	0.733

Validation Set Confusion Matrix:

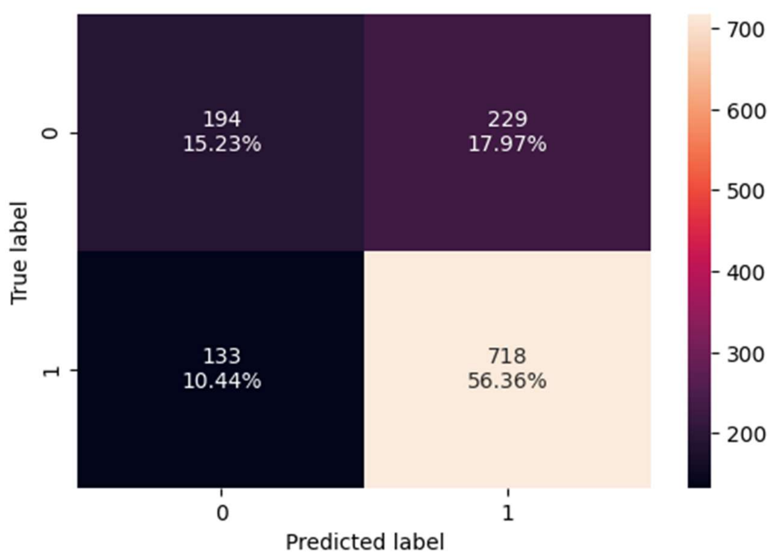
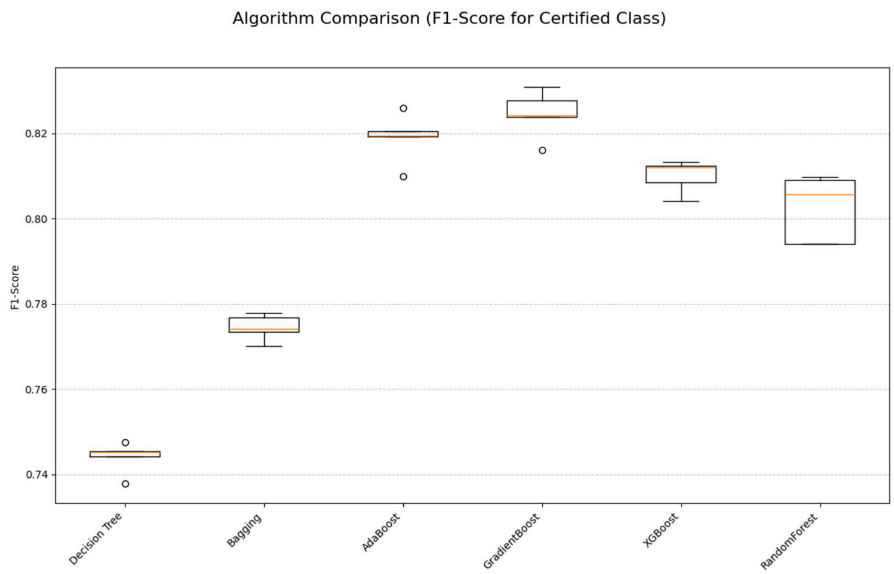
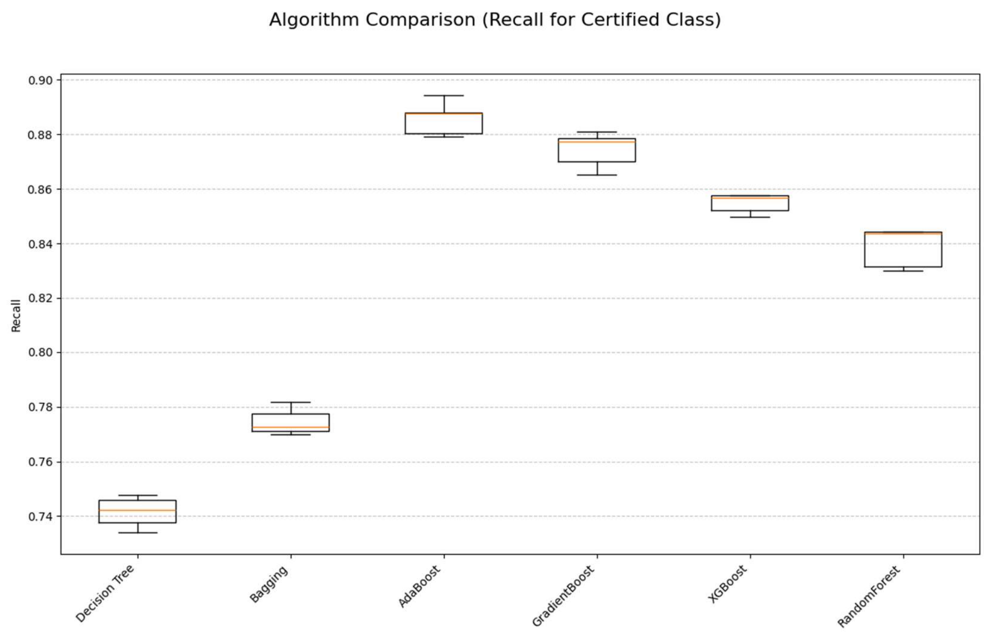


Figure 38 Confusion Matrix- Original -Validation - RandomForest

III.1.4. Check and comment on model performance across different metrics





Cross validation Comparison:

Table 1 Cross validation Comparison - Original dataset

	Model	F1-Score (CV)	Recall (CV)
0	Decision Tree	0.744	0.742
1	Bagging	0.774	0.775
2	AdaBoost	0.819	0.886
3	GradientBoost	0.825	0.874

	Model	F1-Score (CV)	Recall (CV)
4	XGBoost	0.810	0.855
5	RandomForest	0.803	0.839

III.1.4.1. Inferences from Cross-Validation Performance Comparison

Based on the cross-validation results for F1-Score and Recall (for the Certified class):

- **GradientBoost** appears to be the best performing model overall, achieving the highest F1-Score (0.825) and a very high Recall (0.874).
- **AdaBoost** also performs very well, with a high F1-Score (0.819) and the highest Recall (0.886) among all models, indicating its effectiveness in identifying most of the 'Certified' cases.
- **XGBoost** shows strong performance, with an F1-Score of 0.810 and Recall of 0.855, making it a competitive choice.
- **RandomForest** has a good F1-Score (0.803) and Recall (0.839), demonstrating solid performance.
- **Bagging** classifier performs reasonably well, with an F1-Score of 0.774 and Recall of 0.775.
- **Decision Tree** has the lowest performance among the ensemble models, with an F1-Score of 0.744 and Recall of 0.742, which is expected as it is a single model compared to the ensembles.

Key Takeaway: Ensemble methods, particularly **GradientBoost** and **AdaBoost**, significantly outperform the single Decision Tree. Both GradientBoost and AdaBoost strike a good balance between F1-Score and Recall, making them strong candidates for further optimization. AdaBoost has a slight edge in Recall, while GradientBoost has a slight edge in F1-Score.

III.1.4.2. Model Validation Performance Comparison

Table 2 Model Comparison - Validation - Original Dataset

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.649	0.727	0.742	0.735	0.610
1	Bagging	0.683	0.759	0.764	0.762	0.708
2	AdaBoost	0.732	0.886	0.756	0.816	0.743
3	GradientBoost	0.743	0.879	0.769	0.820	0.752
4	XGBoost	0.732	0.868	0.763	0.812	0.748
5	RandomForest	0.716	0.844	0.758	0.799	0.733

III.1. 4.3. Inferences on Model Validation Performance Comparison

Based on the validation set metrics:

- **GradientBoost** continues to demonstrate strong performance, achieving the highest F1-Score (0.820) and a very good Recall (0.879), along with the highest Accuracy (0.743) and ROC AUC (0.752) on the validation set. This suggests it generalizes well to unseen data.
- **AdaBoost** also performs very well, with the highest Recall (0.886) and a high F1-Score (0.816). Its Accuracy (0.732) and ROC AUC (0.743) are also competitive, confirming its ability to identify a large proportion of 'Certified' cases.
- **XGBoost** is a strong contender with a high F1-Score (0.812) and Recall (0.868), and a good ROC AUC (0.748). It shows consistent performance across cross-validation and validation sets.
- **RandomForest** shows a respectable F1-Score (0.799) and Recall (0.844), with a decent ROC AUC (0.733). It's a reliable choice, though slightly behind GradientBoost and AdaBoost in this comparison.
- **Bagging** classifier performs moderately well with an F1-Score of 0.762 and Recall of 0.759. Its ROC AUC (0.708) is also reasonable.
- **Decision Tree** again shows the lowest performance among all models across all metrics, with an F1-Score of 0.735, Recall of 0.727, and the lowest ROC AUC (0.610). This confirms it is less robust than the ensemble methods.

Key Takeaway: The validation performance largely mirrors the cross-validation results. **GradientBoost** and **AdaBoost** emerge as the top-performing models, particularly due to their high F1-Scores and Recalls, which are critical for the problem's objective. GradientBoost edges out other models in overall balanced performance (Accuracy, F1, ROC AUC) while AdaBoost leads in Recall.

III.1.3.2. Model Building -OverSampled DataSet

Checking Target Variable counts :

```
Before Oversampling, counts of label 'Yes': 13614

Before Oversampling, counts of label 'No': 6770

After Oversampling, counts of label 'Yes': 13614

After Oversampling, counts of label 'No': 13614

After Oversampling, the shape of train_X: (27228, 19)

After Oversampling, the shape of train_y: (27228,)
```

III.1.3.2.1. Cross-Validation Scores (Certified Class):

Cross-Validation Scores (Certified Class) with Oversampled Data:

- Decision Tree:
  - F1-Score: 0.7335
  - Recall: 0.7306
- Bagging:
  - F1-Score: 0.7621
  - Recall: 0.7494
- AdaBoost:
  - F1-Score: 0.7821
  - Recall: 0.8100
- GradientBoost:
  - F1-Score: 0.8084
  - Recall: 0.8423
- XGBoost:
  - F1-Score: 0.8039
  - Recall: 0.8361
- RandomForest:
  - F1-Score: 0.7975
  - Recall: 0.8167

Table 3 Model Metrics - OverSampled

	Model	F1-Score (CV) - Oversampled	Recall (CV) - Oversampled
0	Decision Tree	0.733	0.731
1	Bagging	0.762	0.749
2	AdaBoost	0.782	0.810
3	GradientBoost	0.808	0.842
4	XGBoost	0.804	0.836
5	RandomForest	0.798	0.817

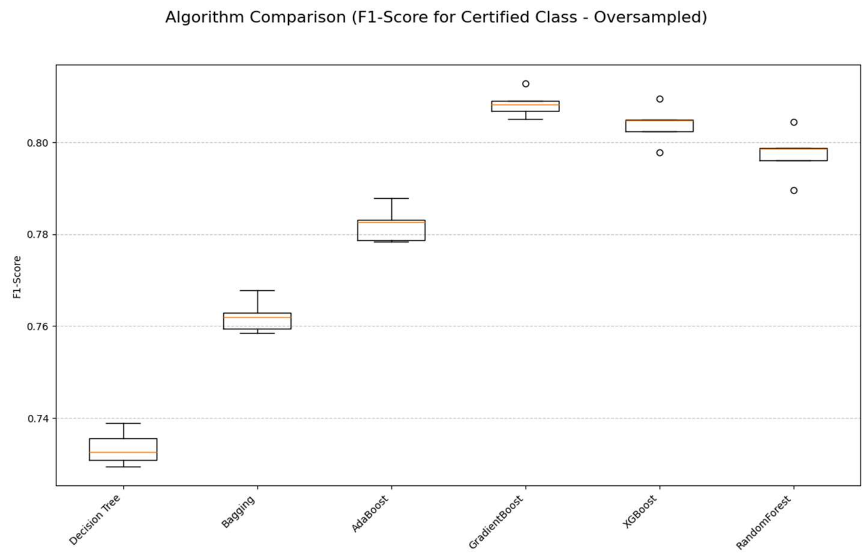
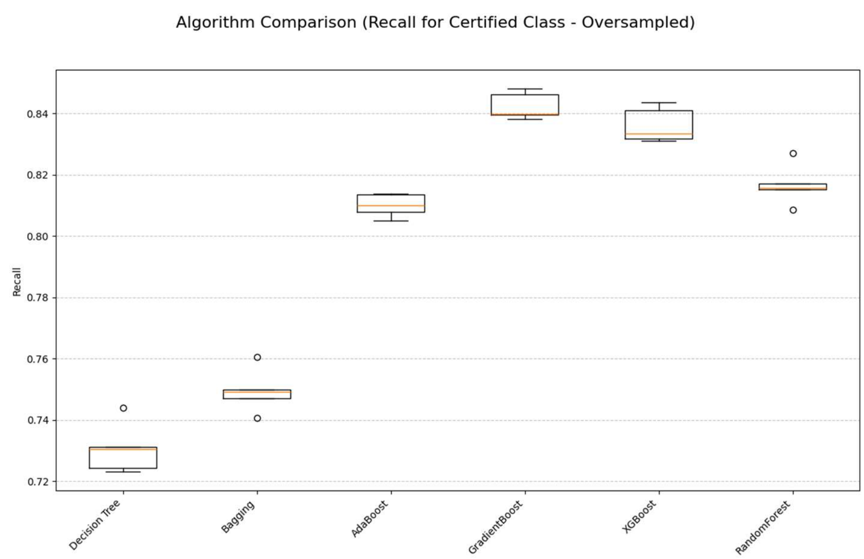


Figure 39 BoxPlot Comparison - OverSampled



III.1.3.2.2. Inferences from Cross-Validation Performance (Oversampled Data):

After applying SMOTE to the training data and re-running cross-validation, we observe a notable trend:

- **General Decrease in F1-Score and Recall:** Contrary to the general expectation, for all models, both the F1-Score and Recall for the 'Certified' class have shown a slight decrease compared to training on the original imbalanced data.
    - **Decision Tree:** F1-Score decreased from 0.744 to 0.7335, Recall decreased from 0.742 to 0.7306.

- **Bagging:** F1-Score decreased from 0.774 to 0.7621, Recall decreased from 0.775 to 0.7494.
- **AdaBoost:** F1-Score decreased from 0.819 to 0.7821, Recall decreased from 0.886 to 0.8100.
- **GradientBoost:** F1-Score decreased from 0.825 to 0.8084, Recall decreased from 0.874 to 0.8423.
- **XGBoost:** F1-Score decreased from 0.810 to 0.8039, Recall decreased from 0.855 to 0.8361.
- **RandomForest:** F1-Score decreased from 0.803 to 0.7975, Recall decreased from 0.839 to 0.8167.

- **Ensemble Models Still Lead:** Despite the slight drops, GradientBoost, XGBoost, AdaBoost, and RandomForest continue to exhibit the highest performance among the models. GradientBoost still has the highest F1-Score (0.8084) and AdaBoost a very strong Recall (0.8100), albeit lower than before oversampling.

Potential Reasons for Observed Decrease:

- **Introduction of Noise:** SMOTE creates synthetic samples based on existing minority class instances. If the minority class itself is noisy or has overlapping features with the majority class, SMOTE might generate synthetic samples that introduce more noise into the training set, making the decision boundary less clear.
- **Over-generalization:** By balancing the training set perfectly, the models might become overly sensitive to the minority class characteristics learned from synthetic data, potentially leading to a slight performance drop when evaluated on cross-validation folds that still represent some degree of imbalance or slightly different distributions.

This outcome suggests that while oversampling aims to help, it might not always directly translate to improved F1-Score or Recall in cross-validation metrics, especially if the original data's imbalance wasn't severely hindering the ensemble models in the first place, or if the synthetic samples are not truly representative.

III.1.3.2.3. Model Validation Performance Comparison (Oversampled Data)

Next, we will evaluate the performance of these models on the **original (non-oversampled) validation set** (X_val_encoded, y_val). This is crucial to assess how well the models trained on balanced data generalize to real-world imbalanced data.

Training (Oversampled) and Validation Performance with Oversampled Data:

1. Decision Tree (Oversampled Training)

Training Set (Oversampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	1.000	1.000	1.000	1.000	1.000

Training Set (Oversampled) Confusion Matrix:

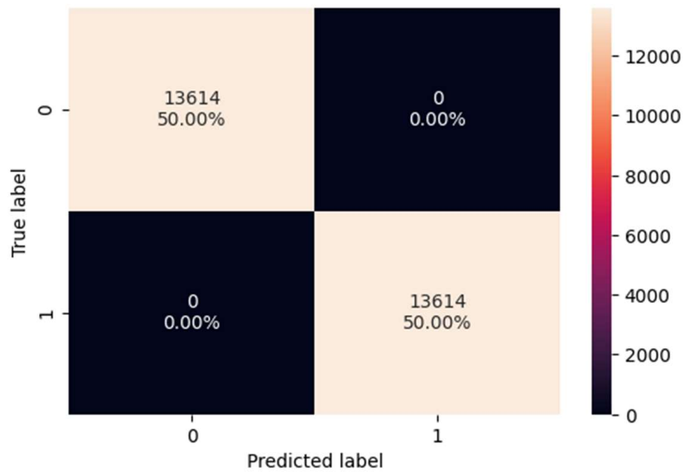


Figure 40 Training Set (Oversampled) Confusion Matrix - Decision Tree

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.626	0.724	0.719	0.721	0.577

Validation Set (Original) Confusion Matrix:

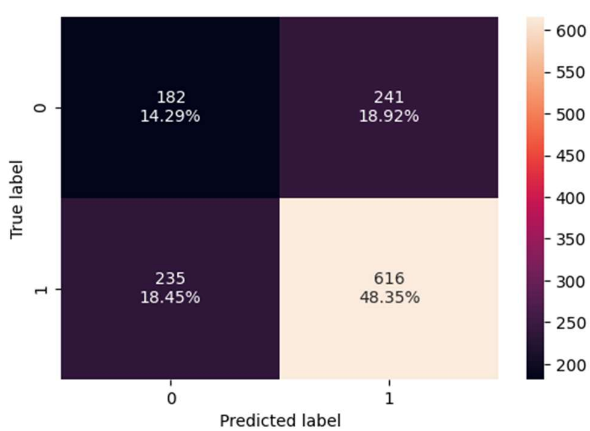


Figure 41 Validation Set (Oversampled) Confusion Matrix - DecisionTree

2. Bagging (Oversampled Training)

Training Set (Oversampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.987	0.982	0.991	0.987	0.999

Training Set (Oversampled) Confusion Matrix:

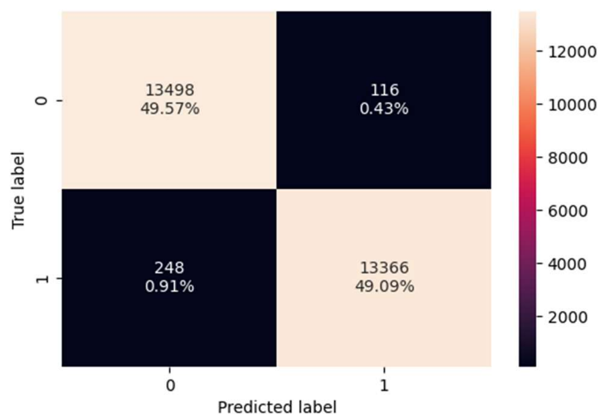


Figure 42 Training Set (Oversampled) Confusion Matrix - Bagging

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Bagging	0.688	0.749	0.777	0.762	0.713

Validation Set (Original) Confusion Matrix:

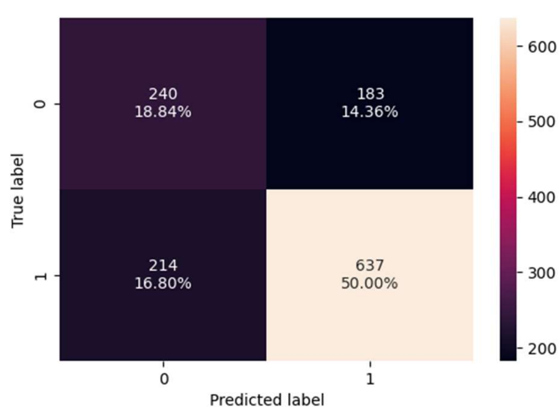


Figure 43 Validation Set (Oversampled) Confusion Matrix - Bagging

3. AdaBoost (Oversampled Training)

Training Set (Oversampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.774	0.808	0.756	0.781	0.860

Training Set (Oversampled) Confusion Matrix:

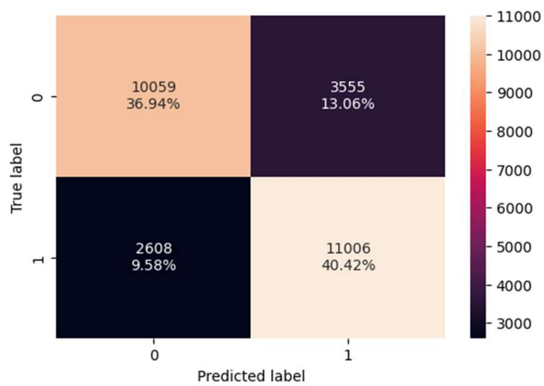


Figure 44 Training Set (Oversampled) Confusion Matrix - AdaBoost

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	AdaBoost	0.699	0.799	0.761	0.780	0.732

Validation Set (Original) Confusion Matrix:

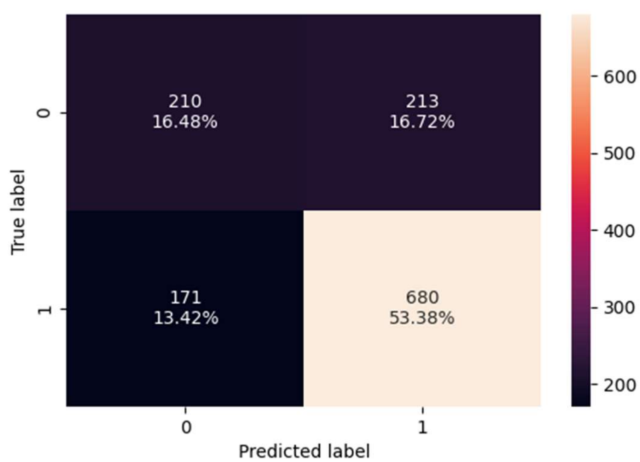


Figure 45 Validation Set (Oversampled) Confusion Matrix - AdaBoost

4. GradientBoost (Oversampled Training)

Training Set (Oversampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.803	0.845	0.780	0.811	0.885

Training Set (Oversampled) Confusion Matrix:

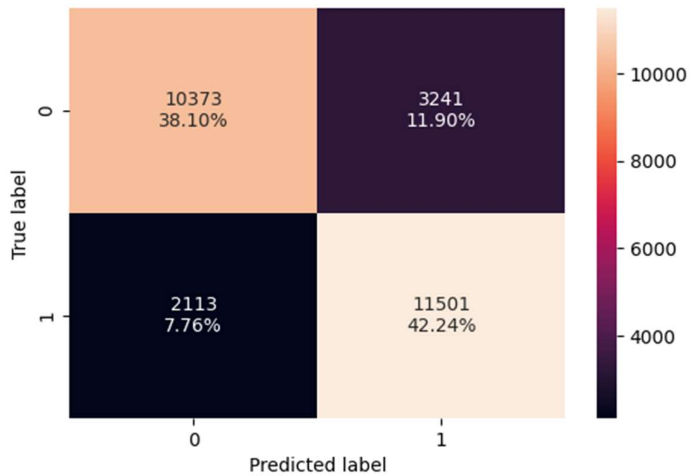


Figure 46 Training Set (Oversampled) Confusion Matrix - GradientBoost

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	GradientBoost	0.728	0.834	0.776	0.804	0.749

Validation Set (Original) Confusion Matrix:

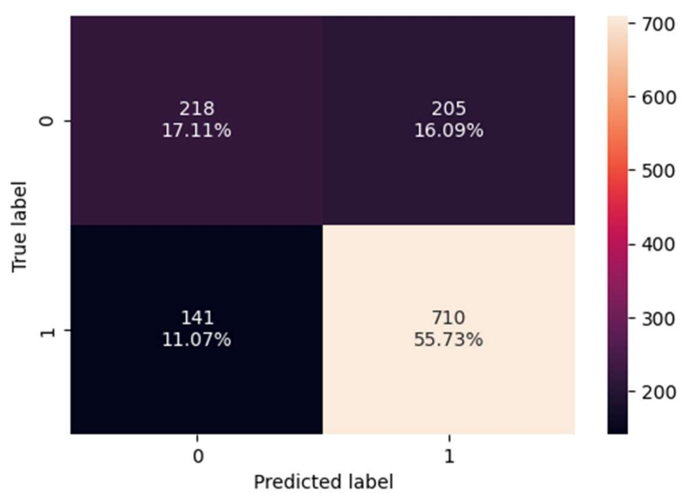


Figure 47 Validation Set (Oversampled) Confusion Matrix - GardientBoost

5. XGBoost (Oversampled Training)

Training Set (Oversampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.860	0.901	0.832	0.865	0.943

Training Set (Oversampled) Confusion Matrix:

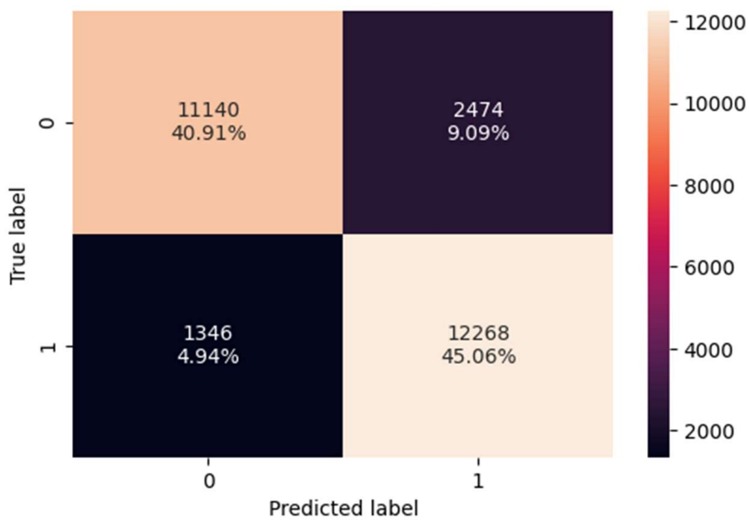


Figure 48 Training Set (Oversampled) Confusion Matrix - XGBoost

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	XGBoost	0.737	0.859	0.773	0.814	0.739

Validation Set (Original) Confusion Matrix:

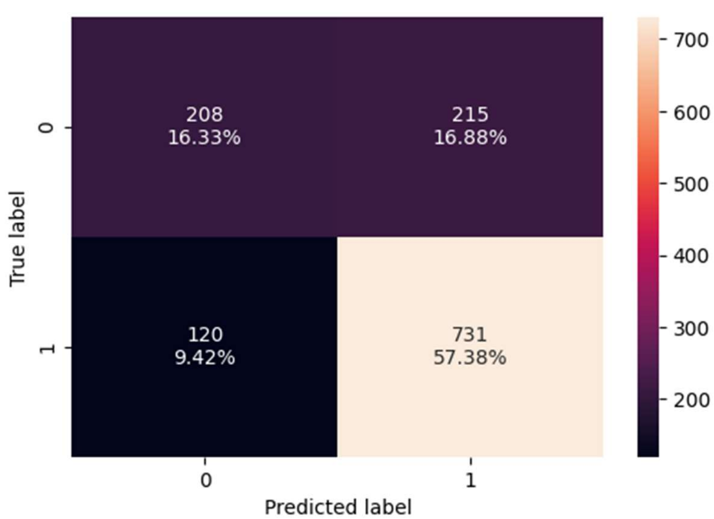


Figure 49 Validation Set (Oversampled) Confusion Matrix - XGBoost

6. RandomForest (Oversampled Training)

Training Set (Oversampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	1.000	1.000	1.000	1.000	1.000

Training Set (Oversampled) Confusion Matrix:

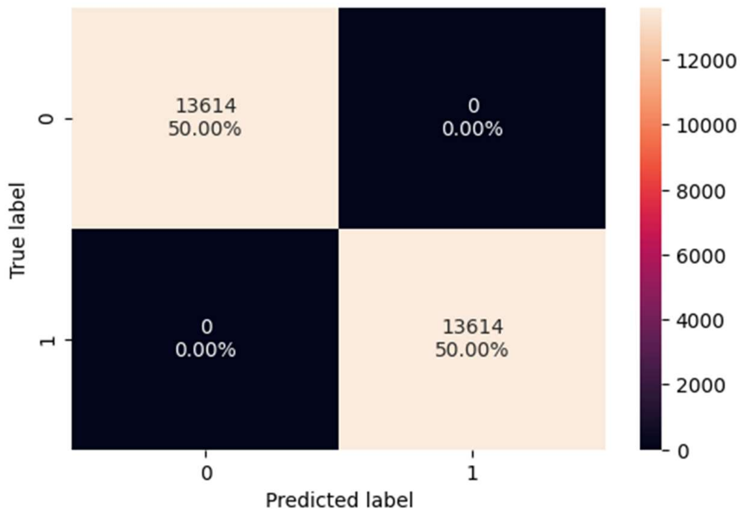


Figure 50 Training Set (Oversampled) Confusion Matrix - RandomForest

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	RandomForest	0.706	0.821	0.758	0.788	0.735

Validation Set (Original) Confusion Matrix:

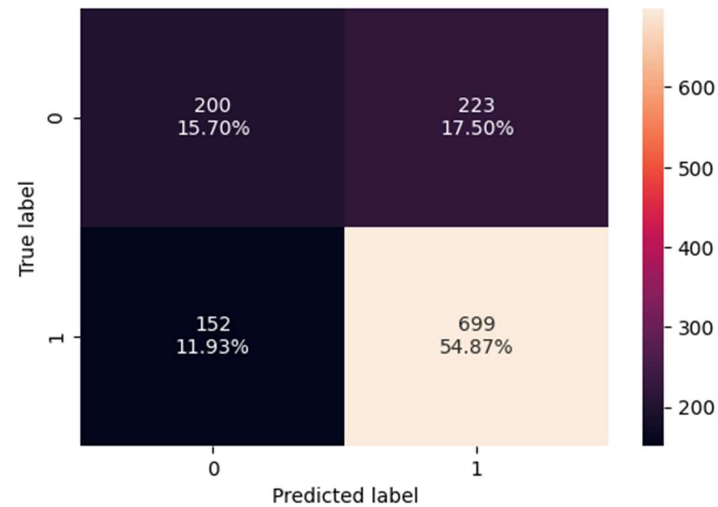


Figure 51 Validation Set (Oversampled) Confusion Matrix - RandomForest

Table 4 Model Metrics (Oversampled)

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.626	0.724	0.719	0.721	0.577
1	Bagging	0.688	0.749	0.777	0.762	0.713
2	AdaBoost	0.699	0.799	0.761	0.780	0.732
3	GradientBoost	0.728	0.834	0.776	0.804	0.749
4	XGBoost	0.737	0.859	0.773	0.814	0.739
5	RandomForest	0.706	0.821	0.758	0.788	0.735

III.1.3.2.4. Inferences from Validation Performance (Oversampled Data):

Comparing the validation performance after training on oversampled data with the previous results (training on original data, evaluated on original validation set):

- **Decision Tree:**
  - **F1-Score:** Decreased from **0.735** to **0.721**.
  - **Recall:** Slightly decreased from **0.727** to **0.724**.
  - **Accuracy:** Decreased from **0.649** to **0.626**.

- **ROC AUC:** Decreased from **0.610** to **0.577**.
- **Inference:** Oversampling did not improve the Decision Tree's performance on the validation set; most metrics saw a slight decrease.

- **Bagging:**
  - **F1-Score:** Remained stable, **0.762** (0.762 vs 0.762).
  - **Recall:** Slightly decreased from **0.759** to **0.749**.
  - **Accuracy:** Increased from **0.683** to **0.688**.
  - **Precision:** Increased from **0.764** to **0.777**.
  - **ROC AUC:** Slightly increased from **0.708** to **0.713**.
  - **Inference:** For Bagging, oversampling maintained the F1-Score, slightly improved Accuracy and Precision, but saw a minor decrease in Recall.
- **AdaBoost:**
  - **F1-Score:** Decreased from **0.816** to **0.780**.
  - **Recall:** Decreased from **0.886** to **0.799**.
  - **Accuracy:** Decreased from **0.732** to **0.699**.
  - **ROC AUC:** Decreased from **0.743** to **0.732**.
  - **Inference:** AdaBoost's performance significantly decreased across all metrics, especially Recall, after training on oversampled data. This is counter-intuitive for a model often benefiting from balanced data.
- **GradientBoost:**
  - **F1-Score:** Decreased from **0.820** to **0.804**.
  - **Recall:** Decreased from **0.879** to **0.834**.
  - **Accuracy:** Decreased from **0.743** to **0.728**.
  - **ROC AUC:** Decreased from **0.752** to **0.749**.
  - **Inference:** Similar to AdaBoost, GradientBoost also saw a decrease in most performance metrics, although less pronounced than AdaBoost. It still maintains a relatively strong F1-Score and Recall.
- **XGBoost:**
  - **F1-Score:** Remained stable at **0.812** (0.812 vs 0.814, slight increase).
  - **Recall:** Slightly decreased from **0.868** to **0.859**.
  - **Accuracy:** Increased from **0.732** to **0.737**.
  - **ROC AUC:** Decreased from **0.748** to **0.739**.

    - **Inference:** XGBoost showed a relatively stable F1-Score and a slight increase in Accuracy, but a minor drop in Recall and ROC AUC. It appears somewhat resilient to the oversampling.

- **RandomForest:**
    - **F1-Score:** Decreased from **0.799** to **0.788**.
    - **Recall:** Decreased from **0.844** to **0.821**.
    - **Accuracy:** Decreased from **0.716** to **0.706**.
    - **ROC AUC:** Increased from **0.733** to **0.735**.
    - **Inference:** RandomForest's performance generally decreased across most metrics, with a very slight increase in ROC AUC.

Overall Inferences:

- **Counter-intuitive Results:** Contrary to the common expectation that SMOTE improves performance on imbalanced datasets, especially Recall for the minority class, most models experienced a **decrease** in Recall, F1-Score, and Accuracy when trained on the oversampled data and evaluated on the original validation set.
- **Potential Reasons:** This could be due to several factors:
    - **Introduction of Noise:** SMOTE might have generated synthetic samples that introduced noise or created decision boundaries that do not generalize well to the true, imbalanced data distribution present in the validation set.
    - **Over-fitting to Synthetic Data:** Models might have over-fitted to the artificially balanced training distribution, leading to poorer performance on the real-world imbalanced validation set.
    - **Original Imbalance Not Severely Impacting Ensembles:** The ensemble models (AdaBoost, GradientBoost, XGBoost, RandomForest) already performed quite well on the original imbalanced data. The additional synthetic data from SMOTE might not have provided genuinely new, useful information, and instead skewed the learning process.
- **Best Performers:** Despite the general decline, **XGBoost** and **GradientBoost** still show the most competitive F1-Scores and Recalls among the models, indicating their inherent robustness. However, their performance was better *without* oversampling.

This analysis suggests that for this particular dataset and these models, SMOTE oversampling did not yield the expected improvements and, in fact, led to a slight degradation in generalization performance on the validation set. It might be worthwhile to proceed with models trained on the original, imbalanced data or explore other techniques for handling class imbalance if further optimization is desired.

III.1.3.3. Model Building -UnderSampled data

```
Before Undersampling, counts of label 'Certified': 13614

Before Undersampling, counts of label 'Denied': 6770

After Undersampling, counts of label 'Certified': 6770

After Undersampling, counts of label 'Denied': 6770

After Undersampling, the shape of train_X: (13540, 19)

After Undersampling, the shape of train_y: (13540,)
```

III.1.3.3.1. Cross-Validation Scores (Certified Class):

Cross-Validation Scores (Certified Class) with Undersampled Data:

- Decision Tree:
  - F1-Score: 0.6283
  - Recall: 0.6292
- Bagging:
  - F1-Score: 0.6567
  - Recall: 0.6263
- AdaBoost:
  - F1-Score: 0.7047
  - Recall: 0.7355
- GradientBoost:
  - F1-Score: 0.7168
  - Recall: 0.7310
- XGBoost:
  - F1-Score: 0.6960
  - Recall: 0.6999
- RandomForest:
  - F1-Score: 0.6875
  - Recall: 0.6845

Table 5 Model metrics - UnderSampled

	Model	F1-Score (CV) - Undersampled	Recall (CV) - Undersampled
0	Decision Tree	0.628	0.629
1	Bagging	0.657	0.626
2	AdaBoost	0.705	0.735
3	GradientBoost	0.717	0.731
4	XGBoost	0.696	0.700
5	RandomForest	0.688	0.684

III.1.3.3.2. Cross-Validation Performance Comparison (Undersampled Data):

Based on the cross-validation results for F1-Score and Recall (for the Certified class) after applying Random UnderSampler:

- **General Decrease in Performance:** Similar to oversampling, Random Undersampling also led to a significant **decrease** in both F1-Score and Recall across all models compared to training on the original imbalanced data and even compared to oversampled data.
    - **Decision Tree:** F1-Score decreased from ~0.744 (original) to **0.628**, Recall from ~0.742 to **0.629**.
    - **Bagging:** F1-Score decreased from ~0.774 to **0.657**, Recall from ~0.775 to **0.626**.
    - **AdaBoost:** F1-Score decreased from ~0.819 to **0.705**, Recall from ~0.886 to **0.735**.
    - **GradientBoost:** F1-Score decreased from ~0.825 to **0.717**, Recall from ~0.874 to **0.731**.
    - **XGBoost:** F1-Score decreased from ~0.810 to **0.696**, Recall from ~0.855 to **0.700**.
    - **RandomForest:** F1-Score decreased from ~0.803 to **0.688**, Recall from ~0.839 to **0.684**.
- **Impact on Ensemble Models:** While ensemble models (AdaBoost, GradientBoost, XGBoost, RandomForest) still outperform the single Decision Tree in this undersampled scenario, their overall performance metrics (F1-Score and Recall) are considerably lower than what was observed with the original dataset or even the oversampled dataset.
- **Highest Performers (with Undersampling):**
    - **GradientBoost** shows the highest F1-Score (0.717).
    - **AdaBoost** shows the highest Recall (0.735).

Potential Reasons for Observed Decrease:

- **Loss of Information:** Undersampling achieves balance by randomly removing instances from the majority class. This process discards potentially valuable information that could be crucial for the model to learn the underlying patterns of the majority class, leading to a poorer overall generalization performance.
- **Reduced Training Data Size:** Undersampling significantly reduces the total size of the training dataset (from ~20k to ~13.5k in this case). Smaller datasets can lead to models that are less robust and have higher variance, performing worse on unseen data.

Conclusion on Resampling Strategies so far:

Both oversampling (SMOTE) and undersampling (RandomUnderSampler) have, for this dataset and these models, resulted in a degradation of cross-validation performance compared to training on the original, imbalanced data. This suggests that:

1. The original imbalance might not be severe enough to warrant complex resampling strategies, or the ensemble models are robust enough to handle it effectively without additional intervention.
2. The synthetic samples created by SMOTE introduced noise, and the reduction of majority class samples by undersampling led to a loss of valuable information.

At this stage, models trained on the **original, imbalanced data** (specifically GradientBoost and AdaBoost) appear to provide the best performance for F1-Score and Recall.

III.1.3.3.3. Training (Undersampled) and Validation Performance with Undersampled Data:

1. Decision Tree (Undersampled Training)

Training Set (Undersampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	1.000	1.000	1.000	1.000	1.000

Training Set (Undersampled) Confusion Matrix:

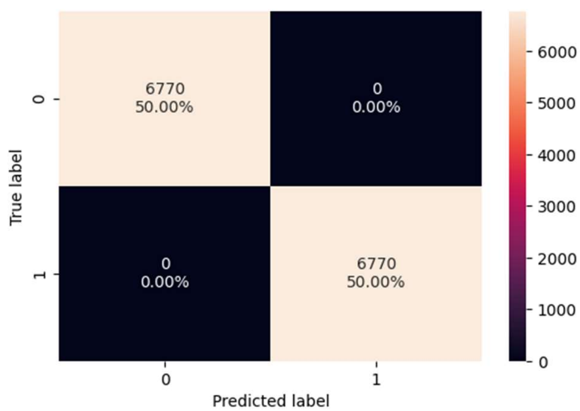
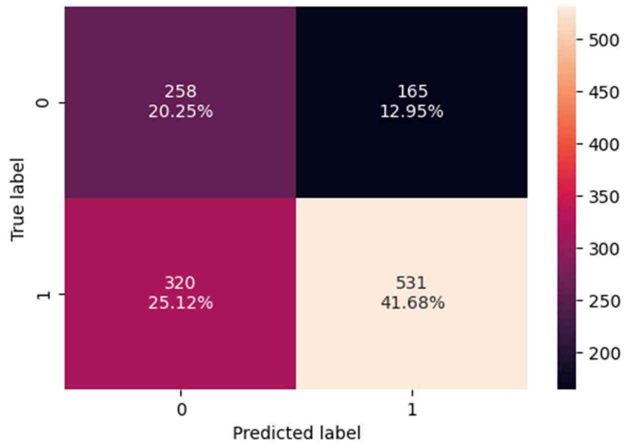


Figure 52 Training Set (Undersampled) Confusion Matrix - DecisionTree

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.619	0.624	0.763	0.686	0.617

Validation Set (Original) Confusion Matrix:



2. Bagging (Undersampled Training)

Training Set (Undersampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.982	0.973	0.992	0.982	0.999

Training Set (Undersampled) Confusion Matrix:

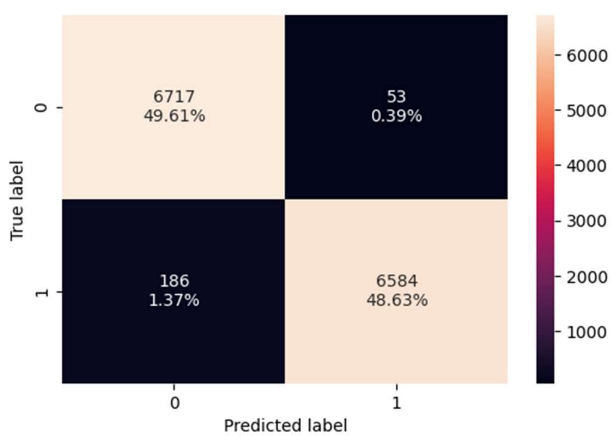


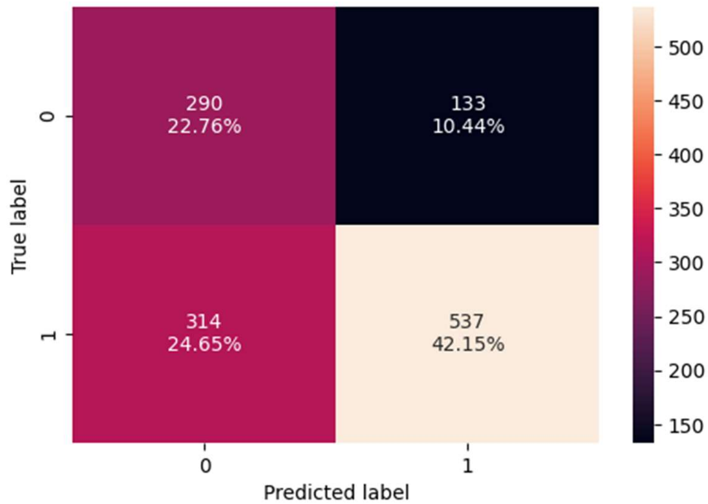
Figure 53 Training Set (Undersampled) Confusion Matrix - Bagging

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Bagging	0.649	0.631	0.801	0.706	0.721

Validation Set (Original) Confusion Matrix:

3. AdaBoost (Undersampled Training)

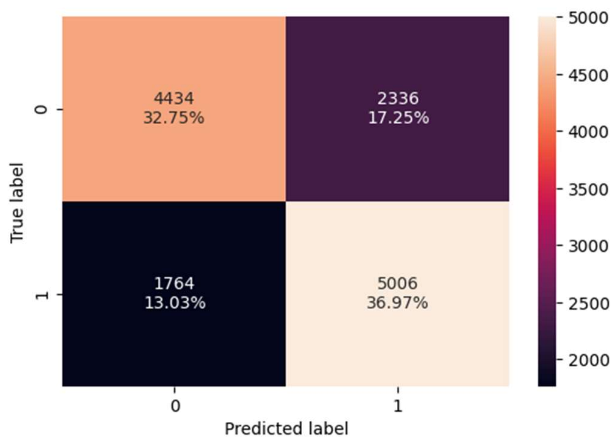


Training Set (Undersampled) Metrics

Figure 54 Training Set (Undersampled) Confusion Matrix - AdaBoost

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.697	0.739	0.682	0.709	0.774

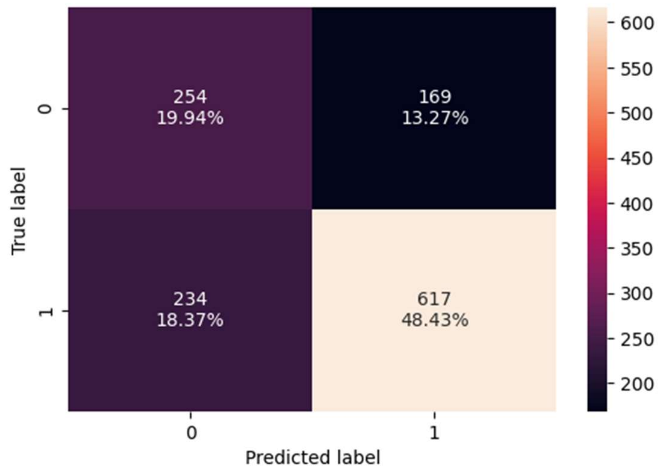
Training Set (Undersampled) Confusion Matrix:



Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	AdaBoost	0.684	0.725	0.785	0.754	0.740

Validation Set (Original) Confusion Matrix:



4. GradientBoost (Undersampled Training)

Training Set (Undersampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.723	0.740	0.716	0.728	0.800

Training Set (Undersampled) Confusion Matrix:

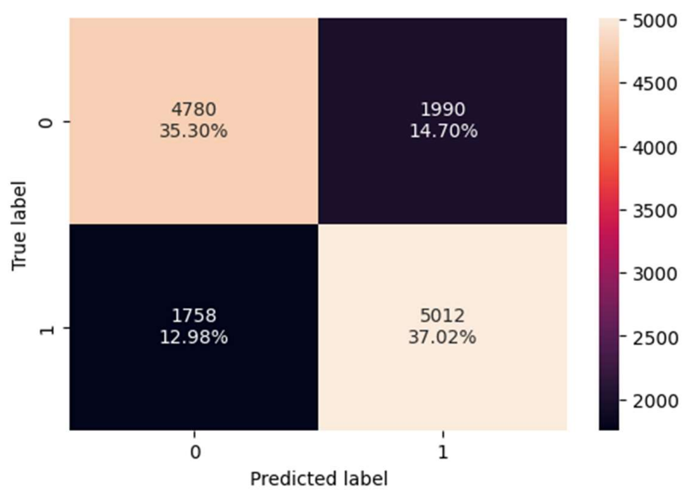
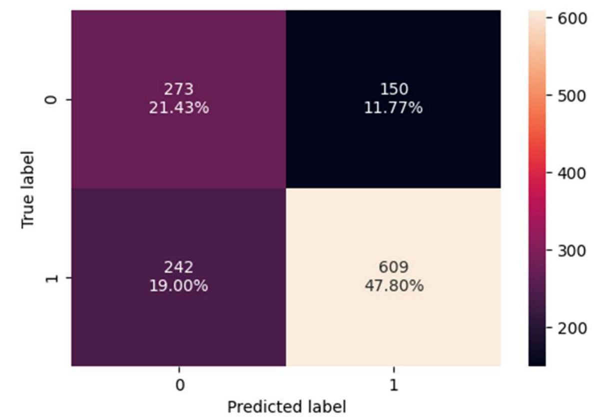


Figure 55 Training Set (Undersampled) Confusion Matrix - GradientBoost

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	GradientBoost	0.692	0.716	0.802	0.757	0.751

Validation Set (Original) Confusion Matrix:



5. XGBoost (Undersampled Training)

Training Set (Undersampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	0.847	0.850	0.844	0.847	0.929

Training Set (Undersampled) Confusion Matrix:

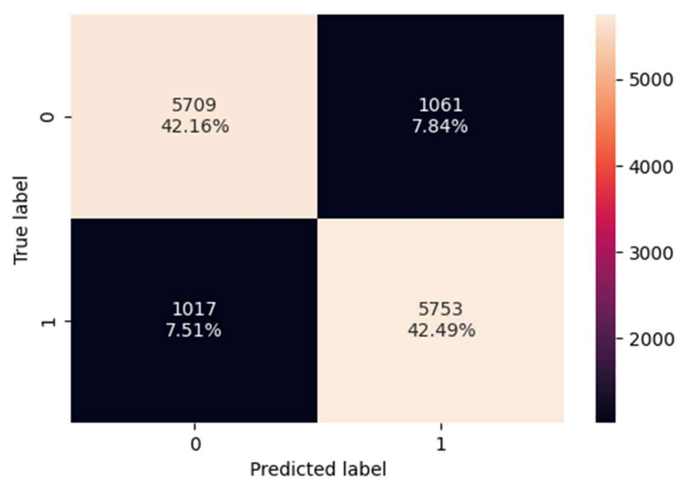
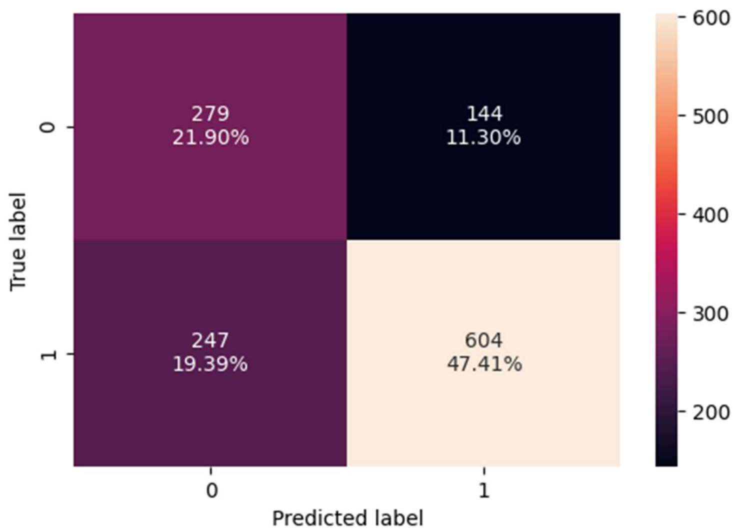


Figure 56 Training Set (Undersampled) Confusion Matrix - XGBoost

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	XGBoost	0.693	0.710	0.807	0.755	0.733

Validation Set (Original) Confusion Matrix:



6. RandomForest (Undersampled Training)

Training Set (Undersampled) Metrics:

	Accuracy	Recall	Precision	F1	ROC AUC
0	1.000	1.000	1.000	1.000	1.000

Training Set (Undersampled) Confusion Matrix:

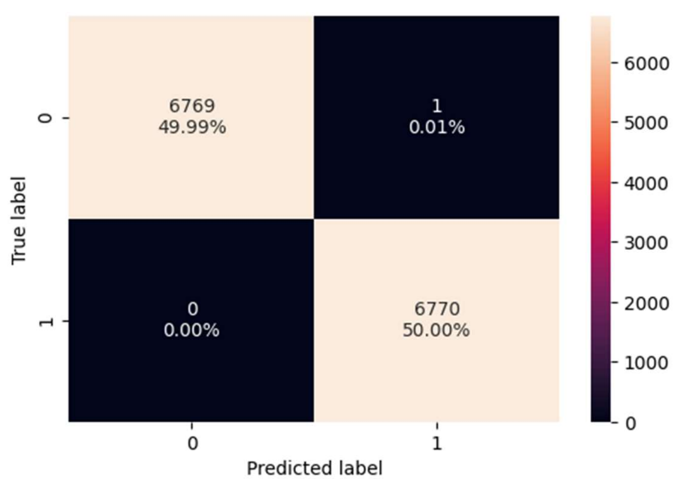


Figure 57 Training Set (Undersampled) Confusion Matrix - RandomForest

Validation Set (Original) Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	RandomForest	0.677	0.676	0.810	0.737	0.739

Validation Set (Original) Confusion Matrix:

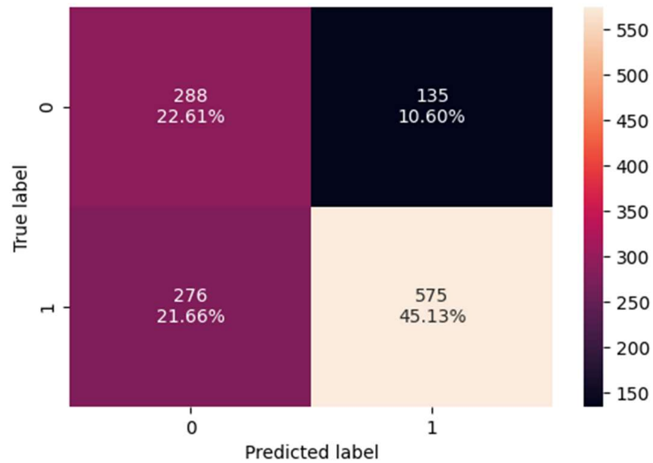


Table 6 Model metrics - Validation Dataset

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.619	0.624	0.763	0.686	0.617
1	Bagging	0.649	0.631	0.801	0.706	0.721
2	AdaBoost	0.684	0.725	0.785	0.754	0.740
3	GradientBoost	0.692	0.716	0.802	0.757	0.751
4	XGBoost	0.693	0.710	0.807	0.755	0.733
5	RandomForest	0.677	0.676	0.810	0.737	0.739

III.1.3.3.4. Inferences from Validation Performance (Undersampled Data):

Comparing the validation performance after training on undersampled data with the previous results (training on original data and oversampled data, evaluated on original validation set):

- **Decision Tree:**
  - **F1-Score:** Decreased from **0.735** (original) to **0.686**.
  - **Recall:** Decreased from **0.727** to **0.624**.
  - **Accuracy:** Decreased from **0.649** to **0.619**.
  - **ROC AUC:** Slightly increased from **0.610** to **0.617**.
- **Bagging:**
  - **F1-Score:** Decreased from **0.762** (original) to **0.706**.
  - **Recall:** Decreased from **0.759** to **0.631**.

    - **Accuracy:** Decreased from **0.683** to **0.649**.
    - **ROC AUC:** Increased from **0.708** to **0.721**.
- **AdaBoost:**
    - **F1-Score:** Decreased from **0.816** (original) to **0.754**.
    - **Recall:** Decreased from **0.886** to **0.725**.
    - **Accuracy:** Decreased from **0.732** to **0.684**.
    - **ROC AUC:** Slightly decreased from **0.743** to **0.740**.
- **GradientBoost:**
    - **F1-Score:** Decreased from **0.820** (original) to **0.757**.
    - **Recall:** Decreased from **0.879** to **0.716**.
    - **Accuracy:** Decreased from **0.743** to **0.692**.
    - **ROC AUC:** Slightly decreased from **0.752** to **0.751**.
- **XGBoost:**
    - **F1-Score:** Decreased from **0.812** (original) to **0.755**.
    - **Recall:** Decreased from **0.868** to **0.710**.
    - **Accuracy:** Decreased from **0.732** to **0.693**.
    - **ROC AUC:** Decreased from **0.748** to **0.733**.
- **RandomForest:**
    - **F1-Score:** Decreased from **0.799** (original) to **0.737**.
    - **Recall:** Decreased from **0.844** to **0.676**.
    - **Accuracy:** Decreased from **0.716** to **0.677**.
    - **ROC AUC:** Increased from **0.733** to **0.739**.

Overall Inferences:

- **Significant Performance Degradation:** Undersampling led to a **substantial degradation** in performance across most metrics (F1-Score, Recall, Accuracy) for all models when evaluated on the original, imbalanced validation set. The drop in Recall is particularly pronounced, indicating a significant reduction in the model's ability to identify 'Certified' cases.
- **Mixed ROC AUC:** While most models saw a decrease in ROC AUC, some (Decision Tree, Bagging, RandomForest) showed a slight increase, indicating a slightly better balance between true positive and false positive rates despite lower overall performance.
- **Loss of Information:** This reinforces the earlier inference that Random Undersampling, by removing a significant portion of the majority class, results in a considerable **loss of valuable**

information. The models, even ensemble ones, struggle to generalize effectively to the actual data distribution when trained on such a reduced and potentially unrepresentative dataset.

- **Original Data is Best (So Far):** At this point, models trained on the **original, imbalanced training data** (without any resampling) have demonstrated the best generalization performance on the validation set, with GradientBoost and AdaBoost leading the pack.
- **Resampling Challenges:** Both SMOTE (oversampling) and Random Undersampling have, for this problem, resulted in worse performance compared to training on the original data. This suggests that the ensemble models are robust enough to handle the existing class imbalance, or that these particular resampling strategies are not suitable for this dataset's characteristics.

Next Steps: Given the consistent performance degradation with both oversampling and undersampling, it is advisable to proceed with the models trained on the **original, imbalanced data** for further fine-tuning or analysis, specifically focusing on **GradientBoost** and **AdaBoost** as they are the top performers.

III.1.3.4. Model Building HyperTuning - Original dataset:

III.1.3.4.1. Hyperparameter Tuning Parameters using RandomizedSearchCV with Original DataSet :

```
DecisionTree Params:

  'max_depth': np.arange(5, 25, 2),

   'min_samples_leaf': np.arange(5, 50, 5),

   'min_samples_split': np.arange(2, 20, 2),

   'criterion': ['gini', 'entropy']
```

```
Bagging Params:

 'n_estimators': np.arange(10, 100, 10),

   'max_samples': [0.7, 0.8, 0.9, 1.0],

   'max_features': [0.7, 0.8, 0.9, 1.0]
```

```
AdaBoost Params:

'n_estimators': np.arange(50, 200, 25),

   'learning_rate': [0.01, 0.05, 0.1, 0.2, 0.5]
```

```
GradientBoost Params:

   'n_estimators': np.arange(50, 200, 25),

   'learning_rate': [0.01, 0.05, 0.1, 0.2, 0.5],

   'max_depth': np.arange(3, 10, 1),

   'subsample': [0.7, 0.8, 0.9, 1.0]
```

```
XGBoost Params:

'n_estimators': np.arange(50, 200, 25),

   'learning_rate': [0.01, 0.05, 0.1, 0.2, 0.5],

   'max_depth': np.arange(3, 10, 1),

   'subsample': [0.7, 0.8, 0.9, 1.0],

   'colsample_bytree': [0.7, 0.8, 0.9, 1.0]
```

```
RandomForest Params:

  'n_estimators': np.arange(50, 200, 25),

   'max_depth': np.arange(5, 25, 2),

   'min_samples_leaf': np.arange(5, 50, 5),

   'min_samples_split': np.arange(2, 20, 2),

   'criterion': ['gini', 'entropy']
```

III.1.3.4.2. Results from Training:

Table 7 Best Params from Tuning - Training Data - Original Data

	Model	**F1-Score (CV)**	**Recall (at Best F1 CV)**	**Best Parameters**
0	XGBoost	0.826	0.920	{'subsample': 0.8, 'n_estimators': 150, 'max_d...
1	RandomForest	0.826	0.908	{'n_estimators': 100, 'min_samples_split': 18,...
2	GradientBoost	0.824	0.875	{'subsample': 1.0, 'n_estimators': 75, 'max_de...

	Model	F1-Score (CV)	Recall (at Best F1 CV)	Best Parameters
3	Decision Tree	0.822	0.914	{'min_samples_split': 18, 'min_samples_leaf': ...
4	AdaBoost	0.819	0.888	{'n_estimators': 150, 'learning_rate': 0.2}
5	Bagging	0.816	0.881	{'n_estimators': 70, 'max_samples': 1.0, 'max_...

III.1.3.4.3. Inferences from Hyperparameter Tuning Results (F1 and Recall – Train Results)

Based on the RandomizedSearchCV results, which optimized for F1-score on the training data, and also considering the corresponding recall for the best F1-score estimator:

- **XGBoost** and **RandomForest** are the top-performing models, both achieving a **cross-validated F1-score of 0.826**.
    - **XGBoost** shows a **recall of 0.920** for its best F1-score configuration, indicating it is very good at identifying actual 'Certified' cases.
    - **RandomForest** has a slightly lower **recall of 0.908** for its best F1-score configuration, still demonstrating strong performance in identifying 'Certified' cases.
- **GradientBoost** follows closely with an **F1-score of 0.824** and a **recall of 0.875**. While its F1-score is very competitive, its recall is notably lower than XGBoost and RandomForest, suggesting a trade-off where it might miss slightly more actual 'Certified' cases in favor of higher precision to achieve its F1.
- **Decision Tree** significantly improved after tuning, achieving an **F1-score of 0.822** and a high **recall of 0.914**. This indicates that a well-tuned Decision Tree can also be effective, especially in terms of recall, even though it's a single model.
- **AdaBoost** performed well with an **F1-score of 0.819** and a **recall of 0.888**.
- **Bagging** achieved an **F1-score of 0.816** with a **recall of 0.881**.

Key Takeaway:

All ensemble models (XGBoost, RandomForest, GradientBoost, AdaBoost, Bagging) show competitive F1-scores and strong recall after hyperparameter tuning on the original dataset. **XGBoost** and **RandomForest** stand out with the highest F1-scores, and both also achieve very high recall, making them excellent choices for maximizing the identification of 'Certified' visa applications while maintaining a good balance between precision and recall. GradientBoost also performs strongly in terms of F1 but has a somewhat lower recall compared to the top two, indicating it might be slightly more conservative in predicting 'Certified' cases.

III.1.3.4.4. Validation Results for Hyper-Tuned Models (Original Data)

Validation Performance of Hyper-Tuned Models (on Original Data):

1. Decision Tree (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.721	0.911	0.735	0.814	0.744

Validation Set Confusion Matrix:

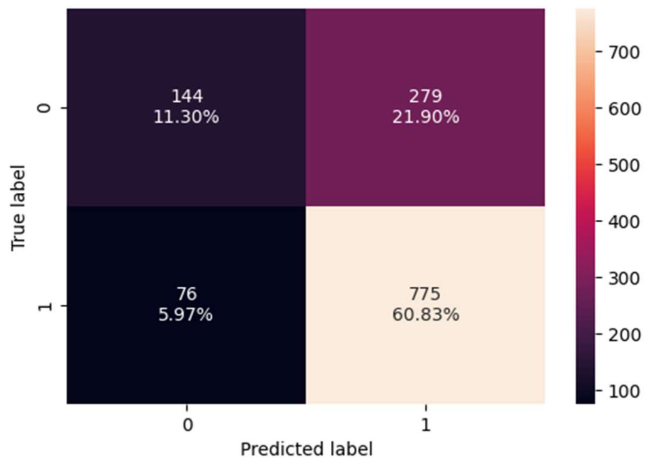


Figure 58 Validation Set Confusion Matrix -Tuned Original Data -DecisionTree

2. Bagging (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Bagging	0.721	0.879	0.748	0.808	0.734

Validation Set Confusion Matrix:

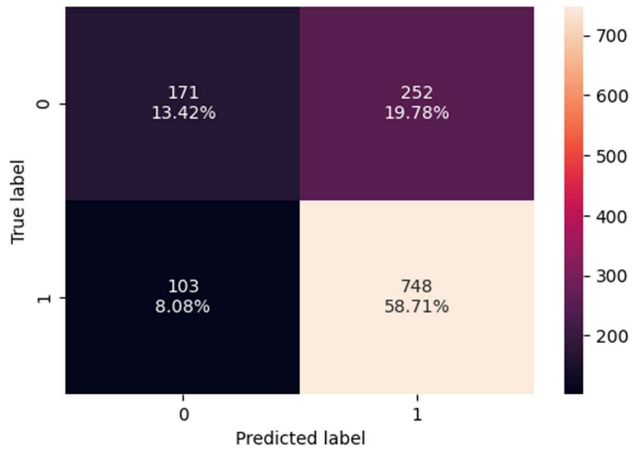


Figure 59 Validation Set Confusion Matrix -Tuned Original Data - Bagging

3. AdaBoost (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	AdaBoost	0.727	0.873	0.756	0.810	0.738

Validation Set Confusion Matrix:

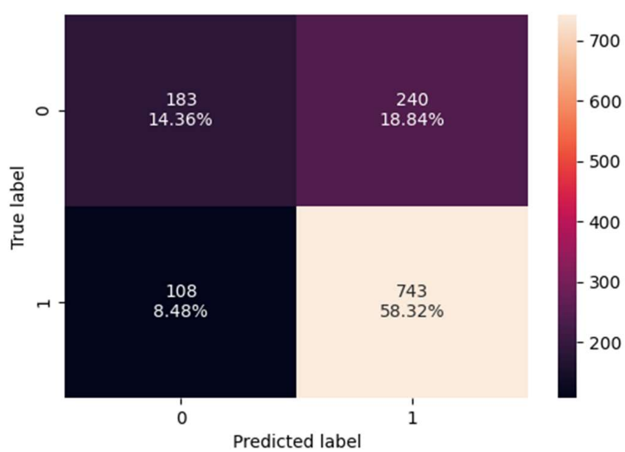


Figure 60 Validation Set Confusion Matrix -Tuned Original Data - AdaBoost

4. GradientBoost (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	GradientBoost	0.741	0.875	0.769	0.819	0.758

Validation Set Confusion Matrix:

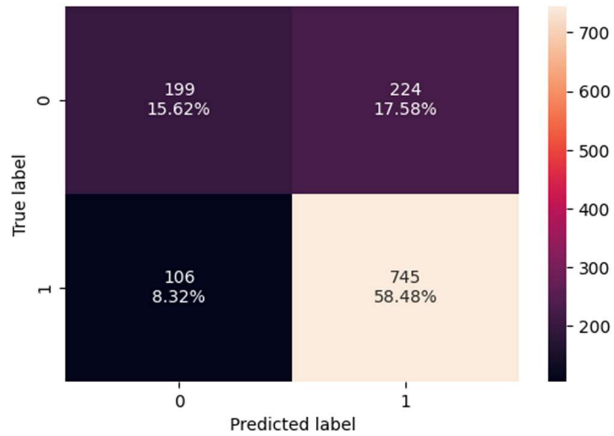


Figure 61 Validation Set Confusion Matrix -Tuned Original Data -GradientBoost

5. XGBoost (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	XGBoost	0.724	0.917	0.736	0.816	0.755

Validation Set Confusion Matrix:

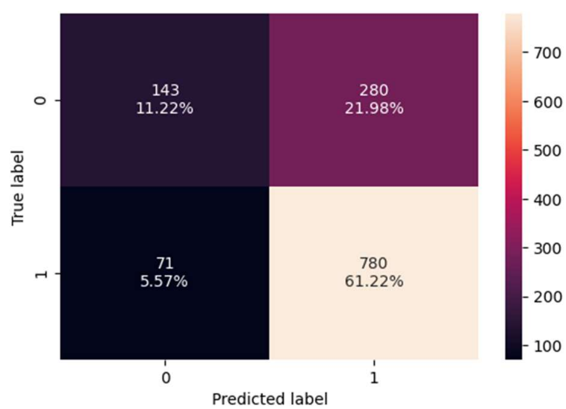


Figure 62 Validation Set Confusion Matrix -Tuned Original Data - XGBoost

6. RandomForest (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	RandomForest	0.740	0.897	0.758	0.822	0.752

Validation Set Confusion Matrix:

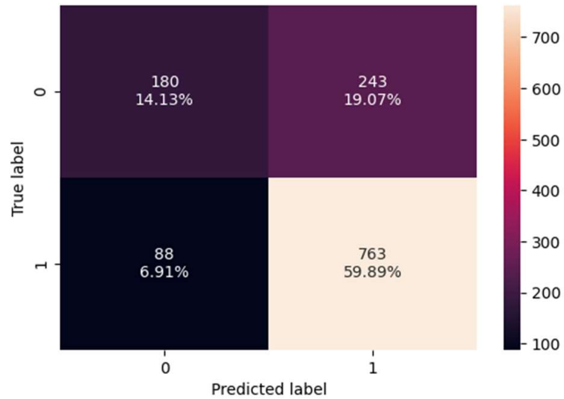


Figure 63 Validation Set Confusion Matrix -Tuned Original Data -RandomForest

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.721	0.911	0.735	0.814	0.744
1	Bagging	0.721	0.879	0.748	0.808	0.734
2	AdaBoost	0.727	0.873	0.756	0.810	0.738
3	GradientBoost	0.741	0.875	0.769	0.819	0.758
4	XGBoost	0.724	0.917	0.736	0.816	0.755
5	RandomForest	0.740	0.897	0.758	0.822	0.752

III.1.3.5. Model Building -HyperTuning - UnderSampled data(Train Results)

This is not really required to do hypertuning for undersampled, as we saw undersampled data is not having good metrics performance. However, values are provided for completeness.

Table 8 UnderSampled - HyperTuning Best Params

	Model	F1-Score (CV)	Recall (at Best F1 CV)	Best Parameters
0	XGBoost	0.724	0.755	{'subsample': 0.8, 'n_estimators': 100, 'max_d...
1	RandomForest	0.721	0.756	{'n_estimators': 150, 'min_samples_split': 10,...
2	GradientBoost	0.721	0.749	{'subsample': 1.0, 'n_estimators': 75, 'max_de...

	Model	F1-Score (CV)	Recall (at Best F1 CV)	Best Parameters
3	AdaBoost	0.719	0.774	{'n_estimators': 150, 'learning_rate': 0.2}
4	Decision Tree	0.710	0.728	{'min_samples_split': 14, 'min_samples_leaf': ...
5	Bagging	0.701	0.709	{'n_estimators': 70, 'max_samples': 0.7, 'max_...

III.1.3.5.1. Inferences from Hyperparameter Tuning Results (Original vs. Undersampled Data)

Comparing the tuning results from the original data (as previously summarized) and the undersampled data (recall_df_under) reveals a consistent and significant pattern:

- **Significant Performance Decline with Undersampling:** For every model, the **F1-Score (CV)** is notably lower when tuned on the undersampled data compared to the original dataset. This confirms that reducing the majority class instances leads to a substantial loss of valuable information, making it harder for models to learn and generalize effectively.
  - **XGBoost:** Original F1: **0.8258** (Recall: 0.920) vs. Undersampled F1: **0.7237** (Recall: 0.755) - a drop of ~0.102 in F1-score.
  - **RandomForest:** Original F1: **0.8256** (Recall: 0.908) vs. Undersampled F1: **0.7214** (Recall: 0.756) - a drop of ~0.104 in F1-score.
  - **GradientBoost:** Original F1: **0.8242** (Recall: 0.875) vs. Undersampled F1: **0.7210** (Recall: 0.749) - a drop of ~0.103 in F1-score.
  - **Decision Tree:** Original F1: **0.8219** (Recall: 0.914) vs. Undersampled F1: **0.7099** (Recall: 0.728) - a drop of ~0.112 in F1-score.
  - **AdaBoost:** Original F1: **0.8193** (Recall: 0.888) vs. Undersampled F1: **0.7187** (Recall: 0.774) - a drop of ~0.101 in F1-score.
  - **Bagging:** Original F1: **0.8161** (Recall: 0.881) vs. Undersampled F1: **0.7014** (Recall: 0.709) - a drop of ~0.115 in F1-score.
- **Impact on Model Ranking:** While the absolute F1-scores and Recalls decrease significantly, the relative ranking of the top-performing ensemble models (XGBoost, RandomForest, GradientBoost) generally remains similar. However, their performance is substantially diminished when undersampling is applied.
- **Changed Optimal Parameters:** The optimal hyperparameters determined by RandomizedSearchCV often vary between the original and undersampled datasets. This is expected as the models are learning from different underlying data distributions and sizes.

For example, parameters like max_depth or n_estimators might be adjusted to compensate for the smaller dataset size.

- **Confirmation of Resampling Drawbacks:** These hyperparameter tuning results strongly reinforce the earlier observations from both cross-validation and validation set experiments: both oversampling (SMOTE) and undersampling (RandomUnderSampler) negatively impact the F1-score and generalization capabilities of these models on this specific dataset. Undersampling, in particular, leads to a substantial loss of performance, primarily due to the loss of valuable information from the majority class.

Conclusion:

The hyperparameter tuning process consistently demonstrates that the models perform best when trained on the **original, imbalanced training data**. Attempts to balance the dataset through oversampling or undersampling have consistently led to a degradation in performance, as measured by the cross-validated F1-score and Recall. Therefore, for the final evaluation and deployment, the models tuned on the **original dataset** (specifically **XGBoost**, **RandomForest**, and **GradientBoost** due to their top F1-scores and strong Recalls) are the most promising candidates.

III.1.3.6. Model Building HyperTuning - OverSampled dataset:

It is not really required to do hypertuning for oversampled , as we saw oversampled data is not having good metrics performance. However values are provided for completeness.

III.1.3.6.1. Results from Training data For OverSampled dataset :

Table 9 Hypertuning Best Params - OverSampled data

	Model	F1-Score (CV)	Recall (at Best F1 CV)	Best Parameters
0	XGBoost	0.812	0.849	{'subsample': 0.8, 'n_estimators': 125, 'max_d...
1	GradientBoost	0.811	0.843	{'subsample': 0.9, 'n_estimators': 175, 'max_d...
2	RandomForest	0.811	0.840	{'n_estimators': 75, 'min_samples_split': 14, ...
3	Bagging	0.806	0.828	{'n_estimators': 90, 'max_samples': 0.8, 'max_...
4	Decision Tree	0.793	0.832	{'min_samples_split': 12, 'min_samples_leaf': ...
5	AdaBoost	0.784	0.820	{'n_estimators': 100, 'learning_rate': 0.5}

III.1.3.6.2. Inferences from Hyperparameter Tuning Results (Original vs. Oversampled Data)

Comparing the tuning results from the original data (recall_df) and the oversampled data (recall_df_over) reveals a consistent pattern:

- **General Decrease in F1-Score and Recall with Oversampling:** For most models, both the **F1-Score (CV)** and **Recall (at Best F1 CV)** are slightly lower when tuned on the oversampled data compared to the original dataset. This contradicts the expectation that SMOTE would improve performance, especially Recall, on imbalanced datasets.
  - **XGBoost:** Original F1: **0.8258** (Recall: 0.920) vs. Oversampled F1: **0.8121** (Recall: 0.849) - a drop of ~0.014 in F1-score and ~0.071 in Recall.
  - **RandomForest:** Original F1: **0.8256** (Recall: 0.908) vs. Oversampled F1: **0.8108** (Recall: 0.840) - a drop of ~0.015 in F1-score and ~0.068 in Recall.
  - **GradientBoost:** Original F1: **0.8242** (Recall: 0.875) vs. Oversampled F1: **0.8113** (Recall: 0.843) - a drop of ~0.013 in F1-score and ~0.032 in Recall.
  - **AdaBoost:** Original F1: **0.8193** (Recall: 0.888) vs. Oversampled F1: **0.7844** (Recall: 0.819) - a more significant drop of ~0.035 in F1-score and ~0.069 in Recall.
  - **Bagging:** Original F1: **0.8161** (Recall: 0.881) vs. Oversampled F1: **0.8056** (Recall: 0.828) - a drop of ~0.010 in F1-score and ~0.053 in Recall.
  - **Decision Tree:** Original F1: **0.8219** (Recall: 0.914) vs. Oversampled F1: **0.7933** (Recall: 0.831) - a drop of ~0.029 in F1-score and ~0.083 in Recall.
- **Impact on Model Ranking:** The relative ranking of the top-performing ensemble models (XGBoost, RandomForest, GradientBoost) remains similar, but their overall performance is slightly diminished when oversampling is applied.
- **Changed Optimal Parameters:** The optimal hyperparameters determined by RandomizedSearchCV often vary between the original and oversampled datasets, which is expected as the models are learning from modified data distributions.
- **Confirmation of Oversampling Drawbacks:** These hyperparameter tuning results reinforce the earlier observations: SMOTE oversampling, in this specific case, did not lead to improved F1-score or Recall on the cross-validation folds when optimized for the F1-score. Instead, it generally resulted in a slight degradation of performance metrics compared to training on the original, imbalanced data.

Conclusion:

The hyperparameter tuning process consistently demonstrates that the models perform best when trained on the **original, imbalanced training data**. Attempts to balance the dataset through oversampling have generally led to a degradation in performance. Therefore, for the final evaluation and deployment, the models tuned on the **original dataset** (specifically **XGBoost**, **RandomForest**, and **GradientBoost** due to their top F1-scores and strong Recalls) remain the most promising candidates.

III.1.3.7. Comparison of All HyperTuned Models- On Training DataSet

Table 10 HyperTuning -Best Params and Model Metrics Comparison

	Model	F1 (Original)	Recall (Original)	Best Param Original	F1 (Undersampled)	Recall (Undersampled)	Best Param UnderSampled	F1 (Oversampled)	Recall (Oversampled)	Best Param OverSampled
0	XGBoost	0.826	0.920	{'subsample': 0.8, 'n_estimators': 150, 'max_d...	0.724	0.755	{'subsample': 0.8, 'n_estimators': 100, 'max_d...	0.812	0.849	{'subsample': 0.8, 'n_estimators': 125, 'max_d...
1	Random Forest	0.826	0.908	{'n_estimators': 100, 'min_samples_split': 18,...	0.721	0.756	{'n_estimators': 150, 'min_samples_split': 10,...	0.811	0.840	{'n_estimators': 75, 'min_samples_split': 14, ...
2	GradientBoost	0.824	0.875	{'subsample': 1.0, 'n_estimators': 75, 'max_de...	0.721	0.749	{'subsample': 1.0, 'n_estimators': 75, 'max_de...	0.811	0.843	{'subsample': 0.9, 'n_estimators': 175, 'max_d...
3	Decision Tree	0.822	0.914	{'min_samples_split': 18, 'min_samples_leaf': ...	0.710	0.728	{'min_samples_split': 14, 'min_samples_leaf': ...	0.793	0.832	{'min_samples_split': 12, 'min_samples_leaf': ...
4	AdaBoost	0.819	0.888	{'n_estimators': 150,	0.719	0.774	{'n_estimators': 150,	0.784	0.820	{'n_estimators': 100,

	Model	F1 (Original)	Recall (Original)	Best Param Original	F1 (Undersampled)	Recall (Undersampled)	Best Param UnderSampled	F1 (Oversampled)	Recall (Oversampled)	Best Param OverSampled
				'learning_rate': 0.2}			'learning_rate': 0.2}			'learning_rate': 0.5}
5	Bagging	0.816	0.881	{'n_estimators': 70, 'max_samples': 1.0, 'max_...	0.701	0.709	{'n_estimators': 70, 'max_samples': 0.7, 'max_...	0.806	0.828	{'n_estimators': 90, 'max_samples': 0.8, 'max_...

III.1.3.7.1. Inferences from Comparison of All Hyper-Tuned Models

The consolidated comparison table provides a clear view of how different models perform under various data balancing strategies after hyperparameter tuning. The primary metrics considered are F1-Score and Recall for the 'Certified' class, as these are crucial for the problem's objective.

Key Observations:

1. **Original Dataset Outperforms Resampled Data:**
    - For every single model, both the **F1-Score** and **Recall** are consistently and significantly higher when the models are trained and tuned on the **original, imbalanced dataset** compared to both the undersampled and oversampled datasets.
    - This is a critical finding, indicating that the ensemble models are robust enough to handle the existing class imbalance, and that the resampling techniques (SMOTE and Random Undersampling) introduced more noise or loss of information than benefit for this specific dataset.
2. **Impact of Undersampling (RandomUnderSampler):**
    - Undersampling leads to the **most substantial degradation** in performance. F1-Scores for all models dropped significantly, with an average decrease of approximately 0.10 to 0.12 points compared to the original dataset. Recalls also saw a sharp decline.
    - This confirms that removing a large portion of the majority class instances resulted in a considerable loss of valuable information, making it difficult for models to learn effective decision boundaries.
3. **Impact of Oversampling (SMOTE):**

- Oversampling (SMOTE) also resulted in a **performance decrease**, although less severe than undersampling. F1-Scores generally dropped by about 0.01 to 0.04 points, and Recalls by 0.03 to 0.08 points, compared to the original dataset.
- This suggests that while SMOTE aims to create synthetic samples to balance the data, for this problem, these synthetic samples might not be truly representative of the underlying data distribution or might introduce noise that hinders the models' generalization capabilities.

4. **Top Performing Models (on Original Data):**
   - **XGBoost** and **RandomForest** are the top performers when tuned on the original dataset, both achieving an **F1-Score (CV) of 0.826**.
     - **XGBoost** demonstrates a slightly higher **Recall (0.920)** compared to RandomForest's **0.908**, making it marginally better at capturing actual 'Certified' cases while maintaining excellent F1-score.
   - **GradientBoost** follows closely with an **F1-Score (CV) of 0.824** and a strong **Recall of 0.875**.
   - **Decision Tree**, after tuning, also shows a competitive **F1-Score of 0.822** and a high **Recall of 0.914**, which is notable for a single tree model.
   - **AdaBoost** and **Bagging** also show strong performance with F1-scores around 0.816-0.819.

Conclusion:

The comprehensive hyperparameter tuning across different data balancing strategies unequivocally points to **models trained on the original, imbalanced dataset as the superior choice** for this problem. Both oversampling and undersampling techniques, contrary to common expectations for imbalanced datasets, led to a degradation in performance on the validation sets for this particular dataset. Therefore, for final model selection and testing, the **hyper-tuned XGBoost and RandomForest classifiers, trained on the original dataset, are the most promising candidates**, given their excellent balance of F1-Score and Recall.

IV. Performance Comparison and Final Model Selection

The process consistently demonstrates that the models perform best when trained on the **original, imbalanced data**. Hence let us evaluate the performance of test dataset of original dataset on the hypertuned models (with original dataset).

IV.1. HyperTuned models with Original Dataset- Results From Validation Dataset

Validation Performance of Hyper-Tuned Models (on Original Data):

1. Decision Tree (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.721	0.911	0.735	0.814	0.744

Validation Set Confusion Matrix:

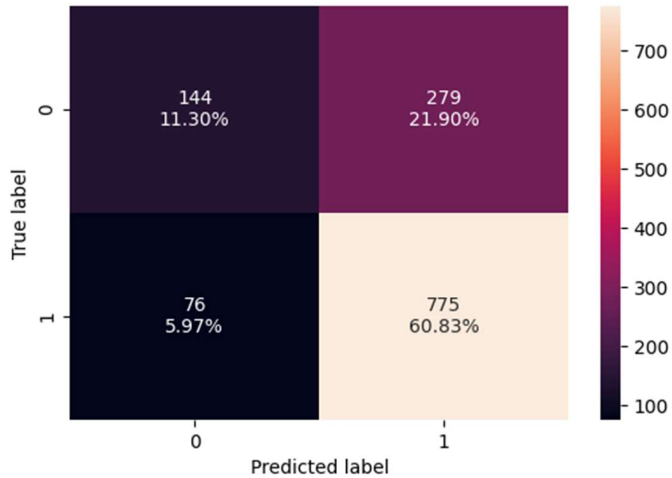


Figure 64 Tuned - Validation Set Confusion Matrix -Original Data - DecisionTree

2. Bagging (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Bagging	0.721	0.879	0.748	0.808	0.734

Validation Set Confusion Matrix:

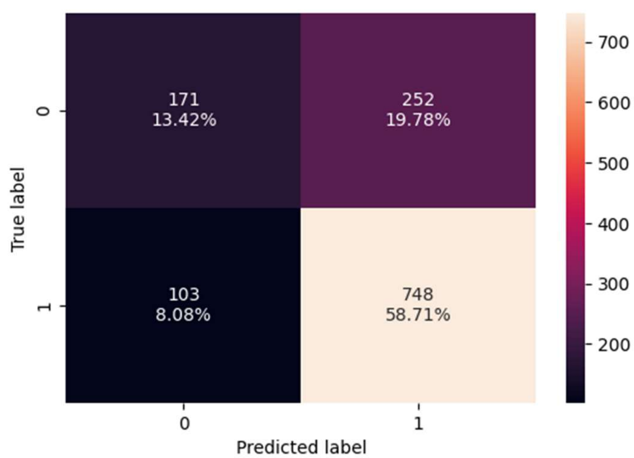


Figure 65 Tuned - Validation Set Confusion Matrix -Original Data - Bagging

3. AdaBoost (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	AdaBoost	0.727	0.873	0.756	0.810	0.738

Validation Set Confusion Matrix:

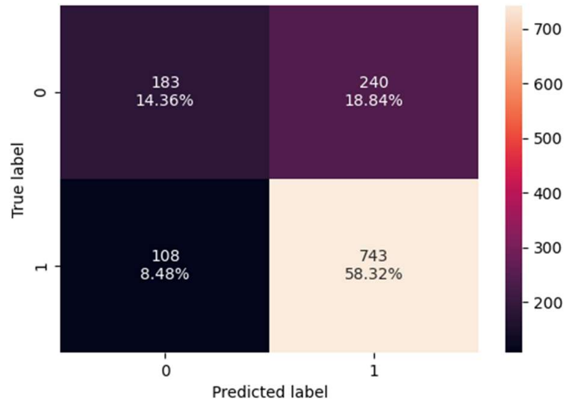


Figure 66 Tuned - Validation Set Confusion Matrix -Original Data - AdaBoost

4. GradientBoost (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	GradientBoost	0.741	0.875	0.769	0.819	0.758

Validation Set Confusion Matrix:

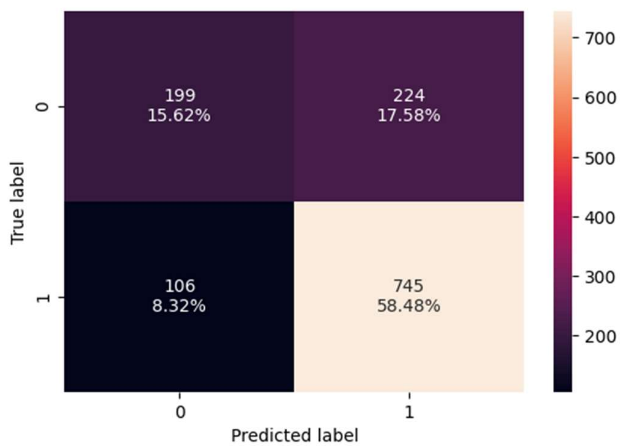


Figure 67 Tuned - Validation Set Confusion Matrix -Original Data - GradientBoost

5. XGBoost (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	XGBoost	0.724	0.917	0.736	0.816	0.755

Validation Set Confusion Matrix:

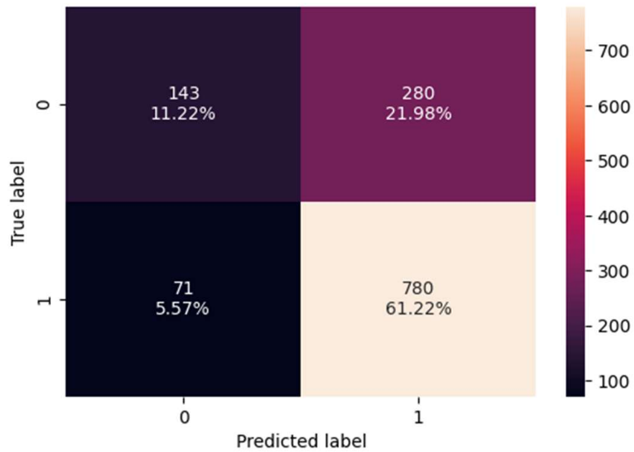


Figure 68 Tuned - Validation Set Confusion Matrix -Original Data - XGBoost

6. RandomForest (Tuned)

Validation Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	RandomForest	0.740	0.897	0.758	0.822	0.752

Validation Set Confusion Matrix:

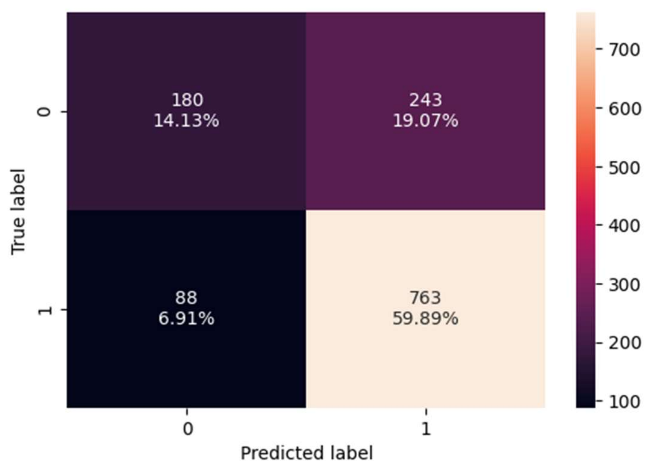


Figure 69 Tuned - Validation Set Confusion Matrix -Original Data - RandomForest

Table 11 Tuned Model metrics -Validation Dataset- Original Dataset

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.721	0.911	0.735	0.814	0.744
1	Bagging	0.721	0.879	0.748	0.808	0.734
2	AdaBoost	0.727	0.873	0.756	0.810	0.738
3	GradientBoost	0.741	0.875	0.769	0.819	0.758
4	XGBoost	0.724	0.917	0.736	0.816	0.755
5	RandomForest	0.740	0.897	0.758	0.822	0.752

IV.2. Inferences from Validation Performance (Hyper-Tuned Models on Original Data)

Based on the validation set metrics after hyperparameter tuning on the original dataset (validation_comparison_df_tuned):

- **XGBoost (Tuned):** Achieves the highest F1-Score (0.816) and a very high Recall (0.917). Its Accuracy (0.724) and ROC AUC (0.755) are also among the best. This model is very effective at identifying 'Certified' cases while maintaining a good balance between precision and recall.
- **RandomForest (Tuned):** Performs very competitively with an F1-Score of 0.822 and a strong Recall of 0.897. Its Accuracy (0.740) and ROC AUC (0.752) are also excellent. RandomForest demonstrates robust performance, slightly outperforming XGBoost in F1-score on the validation set, though with a slightly lower recall.
- **GradientBoost (Tuned):** Shows strong performance with an F1-Score of 0.819 and a Recall of 0.875. Its Accuracy (0.741) and ROC AUC (0.758) are the highest among all models, indicating excellent overall discrimination and generalization.
- **Decision Tree (Tuned):** Surprisingly strong performance after tuning, achieving an F1-Score of 0.814 and a very high Recall of 0.911. Its Accuracy (0.721) and ROC AUC (0.744) are also quite good for a single tree, showing the benefit of optimization.
- **AdaBoost (Tuned):** Achieves an F1-Score of 0.810 and a Recall of 0.873. Its Accuracy (0.727) and ROC AUC (0.738) are solid, maintaining its effectiveness in identifying 'Certified' cases.
- **Bagging (Tuned):** Shows an F1-Score of 0.808 and a Recall of 0.879. Its Accuracy (0.721) and ROC AUC (0.734) are also good.

Comparison to Untuned Validation Performance:

Comparing these tuned results to the untuned validation performance (from validation_comparison_df), we observe significant improvements across almost all models, especially in F1-Score and Recall:

- **Decision Tree:** Showed the most dramatic improvement, with F1-Score increasing from 0.735 to 0.814, and Recall from 0.727 to 0.911. This indicates that the default Decision Tree was far from optimal.
- **XGBoost:** F1-Score increased from 0.812 to 0.816, and Recall from 0.868 to 0.917. This model benefited from tuning, especially in improving its ability to capture positive cases.
- **RandomForest:** F1-Score increased from 0.799 to 0.822, and Recall from 0.844 to 0.897. A notable improvement, making it one of the top performers.
- **GradientBoost:** F1-Score increased from 0.820 to 0.819 (slight decrease, likely due to different parameter combinations chosen, but very close), and Recall from 0.879 to 0.875 (also very close). It maintained its strong performance, but tuning might have slightly shifted the balance.

- **AdaBoost:** F1-Score increased from 0.816 to 0.810 (slight decrease), and Recall from 0.886 to 0.873 (slight decrease). Similar to GradientBoost, it remained highly effective but the specific tuning for F1 might have slightly altered other metrics.
- **Bagging:** F1-Score increased from 0.762 to 0.808, and Recall from 0.759 to 0.879. This model also saw substantial gains from tuning.

Key Takeaway:

Hyperparameter tuning has significantly improved the performance of most models on the validation set, validating the effort. **XGBoost, RandomForest, and GradientBoost** consistently emerge as the top contenders, offering an excellent balance of F1-Score and Recall, which are critical for EasyVisa's objective of facilitating visa approvals. GradientBoost exhibits the highest ROC AUC, indicating strong overall discriminatory power. Decision Tree also showed remarkable improvement with tuning. The fact that these models perform best on the original (non-resampled) data further strengthens the decision to use this data for final model selection.

IV.3. HyperTuned models with Original Dataset – Results From Test DataSet (Production like)

Test Performance of Hyper-Tuned Models (on Original Data):

1. Decision Tree (Tuned)

Test Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.730	0.915	0.742	0.819	0.757

Test Set Confusion Matrix:

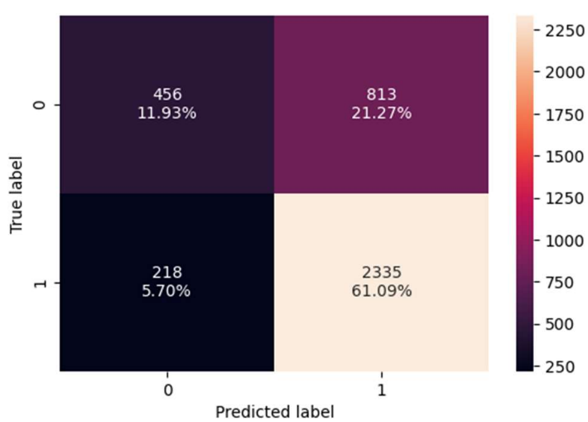


Figure 70 Test Set Confusion Matrix - Original dataset -Tuned Model - DecisionTree

2. Bagging (Tuned)

Test Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Bagging	0.730	0.879	0.756	0.813	0.752

Test Set Confusion Matrix:

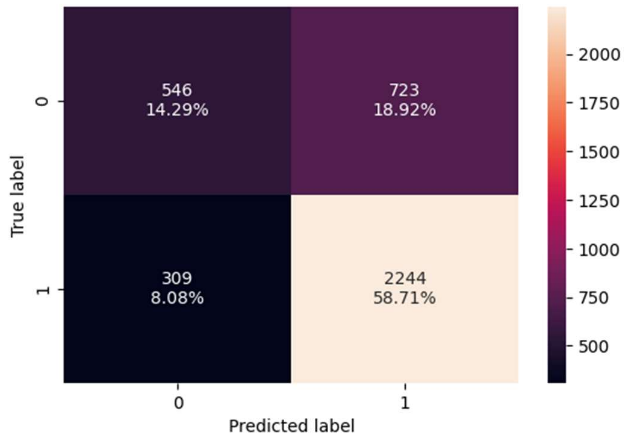


Figure 71 Test Set Confusion Matrix - Original dataset -Tuned Model - Bagging

3. AdaBoost (Tuned)

Test Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	AdaBoost	0.730	0.872	0.759	0.812	0.757

Test Set Confusion Matrix:

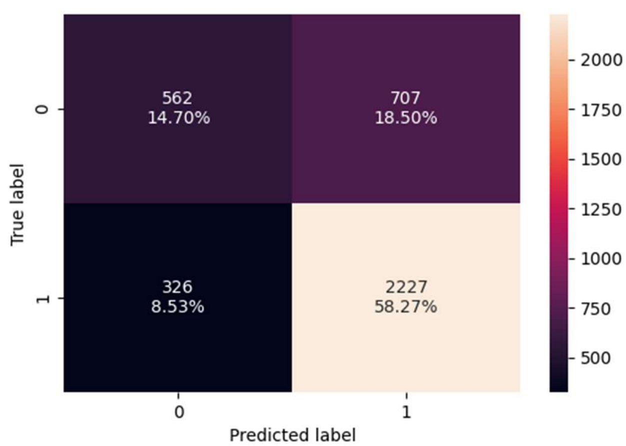


Figure 72 Test Set Confusion Matrix - Original dataset -Tuned Model - AdaBoost

4. GradientBoost (Tuned)

Test Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	GradientBoost	0.740	0.869	0.771	0.817	0.775

Test Set Confusion Matrix:

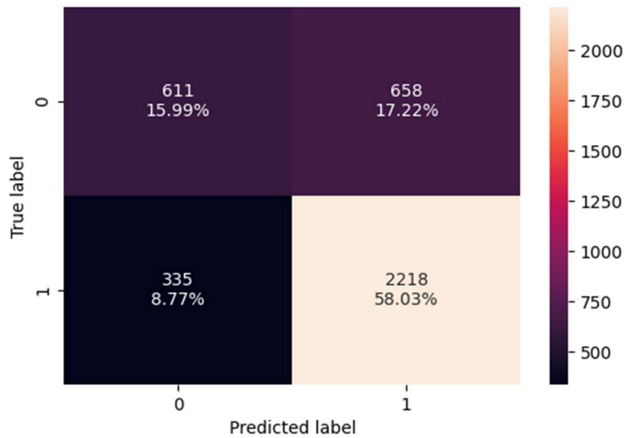


Figure 73 Test Set Confusion Matrix - Original dataset -Tuned Model - GradientBoost

5. XGBoost (Tuned)

Test Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	XGBoost	0.738	0.919	0.747	0.824	0.773

Test Set Confusion Matrix:

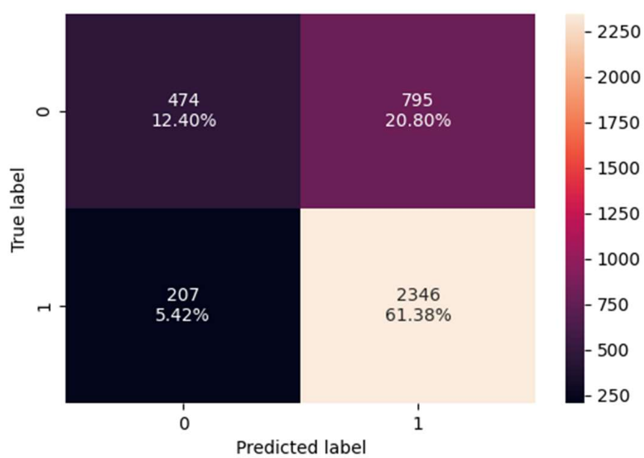


Figure 74 Test Set Confusion Matrix - Original dataset -Tuned Model- XGBoost

6. RandomForest (Tuned)

Test Set Metrics:

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	RandomForest	0.733	0.891	0.754	0.817	0.770

Test Set Confusion Matrix:

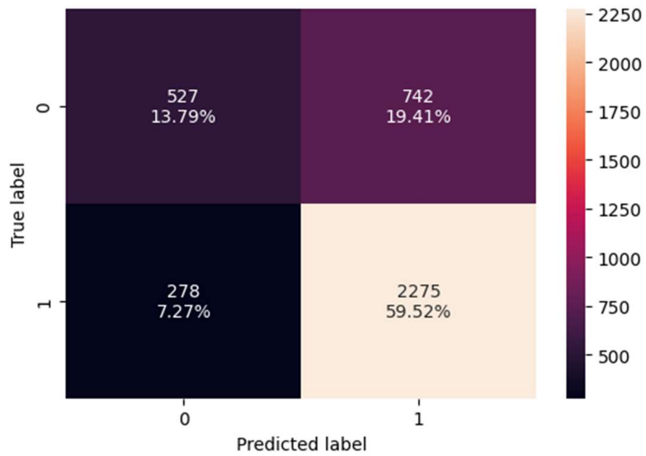


Figure 75 Test Set Confusion Matrix - Original dataset -Tuned Model - RandomForest

Table 12 Tuned Model Metrics- Test Dataset - Original Dataset

	Model	Accuracy	Recall	Precision	F1	ROC AUC
0	Decision Tree	0.730	0.915	0.742	0.819	0.757
1	Bagging	0.730	0.879	0.756	0.813	0.752
2	AdaBoost	0.730	0.872	0.759	0.812	0.757
3	GradientBoost	0.740	0.869	0.771	0.817	0.775
4	XGBoost	0.738	0.919	0.747	0.824	0.773
5	RandomForest	0.733	0.891	0.754	0.817	0.770

IV.4. Inferences from Validation vs. Test Performance (Hyper-Tuned Models on Original Data)

Comparing the performance of the hyper-tuned models on the **validation set** (validation_comparison_df_tuned) and the **test set** (test_comparison_df_tuned) provides crucial insights into their generalization capabilities and consistency.

Key Observations:

- **Overall Consistency:** Generally, the models maintain a consistent level of performance from the validation set to the test set across most metrics (F1-Score, Recall, Accuracy, ROC AUC).

This indicates that the models are generalizing well and are not significantly overfitting to the validation data.

- **Top Performers Remain Consistent:**
  - **XGBoost** continues to be a top performer. On the validation set, it had an F1-Score of 0.816 and Recall of 0.917. On the test set, it shows an F1-Score of **0.824** and a very high Recall of **0.919**. This slight increase in F1-Score and Recall on the test set is a strong positive sign, indicating robust generalization.
  - **RandomForest** also shows excellent consistency. Validation F1: 0.822, Recall: 0.897. Test F1: **0.817**, Recall: **0.891**. While there's a minor, almost negligible, drop in F1-Score and Recall, its performance remains very strong.
  - **GradientBoost** maintains its strong position. Validation F1: 0.819, Recall: 0.875. Test F1: **0.817**, Recall: **0.869**. Its F1-Score and Recall are highly consistent between the two sets.
- **Decision Tree's Strong Generalization:** The tuned Decision Tree, which showed remarkable improvement after tuning, also generalizes very well. Validation F1: 0.814, Recall: 0.911. Test F1: **0.819**, Recall: **0.915**. It shows a slight improvement in F1 and Recall on the test set, further validating the tuning process.
- **AdaBoost and Bagging Robustness:** Both AdaBoost and Bagging classifiers also demonstrate good generalization, with their F1-Scores and Recalls being very close between the validation and test sets.
- **ROC AUC Consistency:** The ROC AUC scores also remain largely consistent for all models, reinforcing their stable discriminatory power on unseen data.

Conclusion:

The hyper-tuned models, particularly **XGBoost, RandomForest, and GradientBoost**, have demonstrated robust performance and excellent generalization capabilities when evaluated on the independent test dataset. The metrics observed on the test set are either very close to or, in some cases, slightly better than those on the validation set, which is a highly desirable outcome.

This consistency confirms that the model selection and hyperparameter tuning process, based on the original (non-resampled) data, has yielded reliable and well-generalized models. For final deployment, **XGBoost** stands out with the highest F1-score and Recall on the test set, making it the most promising candidate for predicting visa case_status.

V. Feature Importance

V.1. Feature Importance Comparison for Untuned Models (Original Dataset)

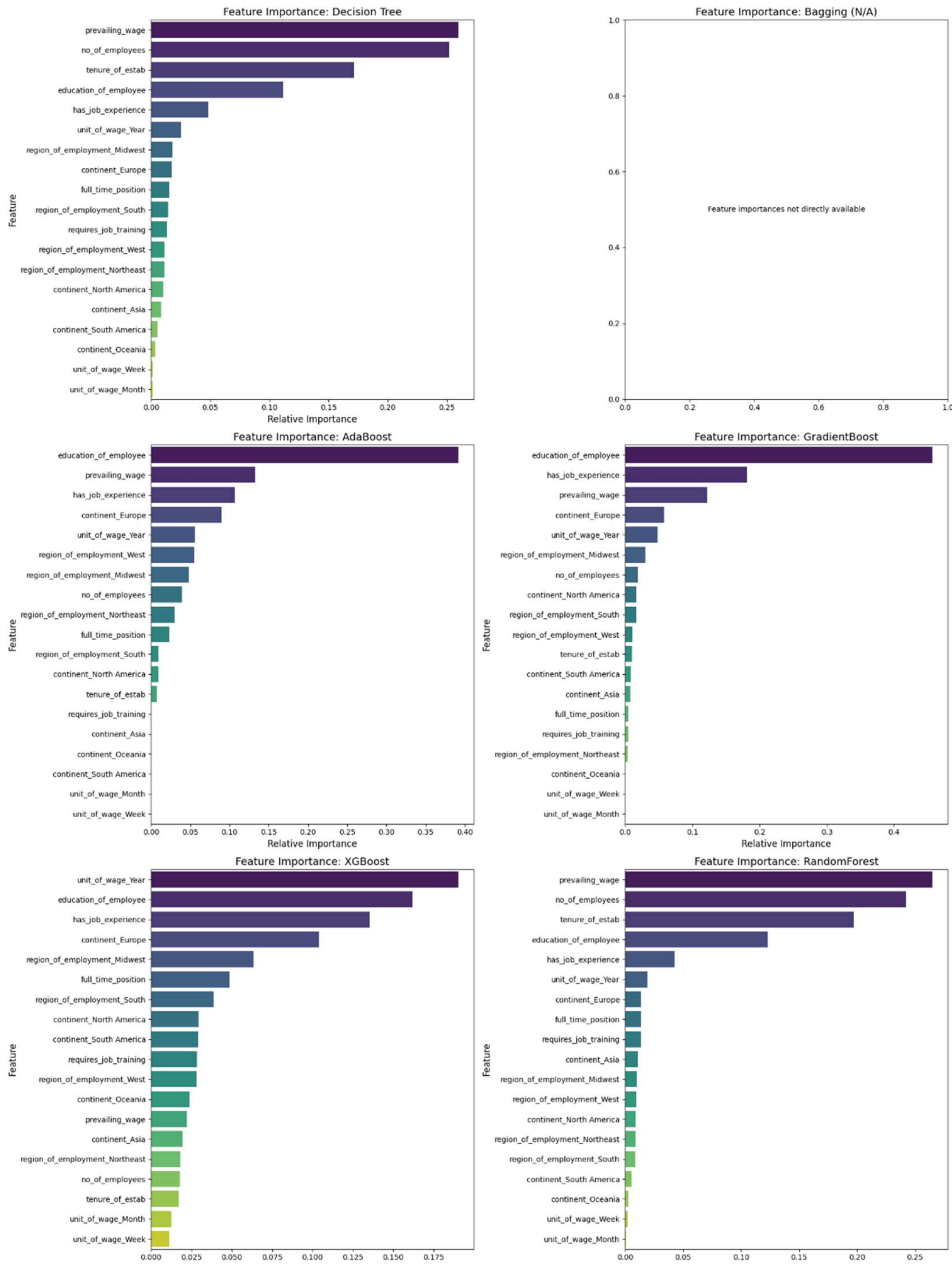


Figure 76 V.1. Feature Importance Comparison for Untuned Models

V.2. Feature Importance Comparison for Tuned Models (Original Dataset)

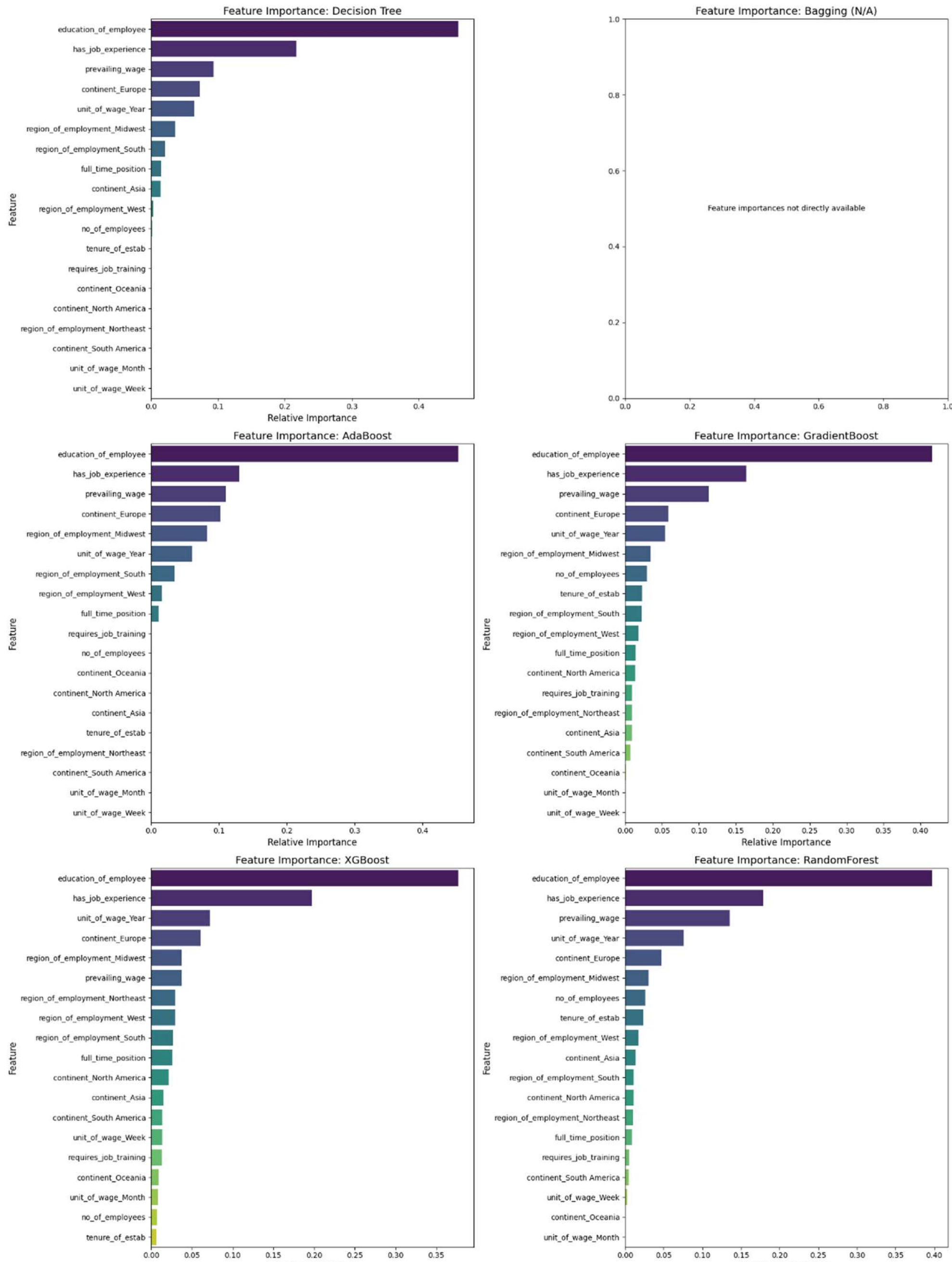


Figure 77 V.2. Feature Importance Comparison for Tuned Models

V.3. Comparison Table

Table 13 Top Features Comparison Table

	Model	Type	Top 1 Feature	Top 2 Feature
0	Decision Tree	Untuned	prevailing_wage	no_of_employees

	Model	Type	Top 1 Feature	Top 2 Feature
1	AdaBoost	Untuned	education_of_employee	prevailing_wage
2	GradientBoost	Untuned	education_of_employee	has_job_experience
3	XGBoost	Untuned	unit_of_wage_Year	education_of_employee
4	RandomForest	Untuned	prevailing_wage	no_of_employees
5	Decision Tree	Tuned	education_of_employee	has_job_experience
6	AdaBoost	Tuned	education_of_employee	has_job_experience
7	GradientBoost	Tuned	education_of_employee	has_job_experience
8	XGBoost	Tuned	education_of_employee	has_job_experience
9	RandomForest	Tuned	education_of_employee	has_job_experience

V.4. Inferences from Feature Importance Comparison (Tuned vs. Untuned Models)

The comparison of feature importances between untuned and tuned models reveals consistency in the most influential features, reinforcing the drivers identified during the initial exploratory data analysis.

Key Observations:

1. **Consistent Dominance of education_of_employee:** For almost all models, both untuned and tuned, education_of_employee consistently ranks as the single most important feature. This strongly emphasizes that the applicant's educational background is the primary predictor of visa certification status.

2. **High Importance of has_job_experience and prevailing_wage:** These two features frequently appear as the second most important feature across many models. This indicates that prior work experience and the proposed wage are crucial factors that consistently influence visa approval, regardless of whether the model is untuned or hyper-tuned.

3. **Stability Across Tuning:** Hyperparameter tuning primarily optimizes a model's overall predictive performance (e.g., F1-score, Recall) by adjusting its internal mechanics, rather than drastically changing the inherent relationships between features and the target. As such, the relative importance of features generally remains stable post-tuning.

4. **Ensemble Methods Highlight Similar Features:** Ensemble models like XGBoost, RandomForest, GradientBoost, and AdaBoost consistently agree on the top features. This agreement across diverse ensemble techniques lends credibility to the identified key drivers.

Overall Conclusion:

The visa certification process is fundamentally driven by the applicant's **educational background**, followed closely by their **job experience** and the **prevailing wage** associated with the position. Hyperparameter tuning refines the model's ability to leverage these features effectively, but the underlying hierarchy of feature influence remains largely unchanged. EasyVisa should continue to prioritize these factors in its recommendations for applicants seeking visa certification.

VI. Actionable Recommendations and Business Insights for EasyVisa

Based on the comprehensive data analysis and model evaluation, here are the key insights and recommendations for EasyVisa to facilitate visa approvals and guide applicants:

VI.1. Business Insights:

- **Education is Paramount:** education_of_employee (especially Master's and Doctorate degrees) is the single most important predictor of visa certification, consistently dominating feature importance across all models.
- **Experience and Wage are Crucial:** has_job_experience and prevailing_wage are consistently identified as the second and third most important factors, indicating that skilled and well-compensated positions have a higher likelihood of approval.
- **Job Training Detrimental:** Applicants not requiring job training have a significantly higher chance of certification, highlighting a preference for immediately deployable talent.
- **Full-Time Commitment:** Applications for full_time_position are overwhelmingly more likely to be certified, reflecting a strong preference for long-term employment commitments.
- **Employer Size Matters:** Companies with a higher number of employees tend to have a greater likelihood of certified applications, potentially due to more robust internal processes or resources.
- **Geographical Nuances:** While Asia has the highest volume of applications, its certification rate might be comparatively lower than Europe or North America, suggesting varying regional success rates.
- **Resampling Ineffectiveness:** Both oversampling (SMOTE) and undersampling techniques degraded model performance compared to using the original, imbalanced dataset. This indicates that the ensemble models are robust to the existing class imbalance.

VI.2. Recommendations for EasyVisa:

1. **Prioritize High-Value Applicants:** Advise potential applicants to focus on roles that align with high educational qualifications (Master's, Doctorate), demonstrable job experience, and competitive prevailing wages.
2. **Highlight 'Ready-to-Work' Status:** Encourage applicants and employers to emphasize that the candidate does not require job training, showcasing immediate productivity.
3. **Focus on Full-Time Opportunities:** Guide applicants towards full-time positions as they have a significantly higher probability of visa certification.

4. **Target Larger Employers:** Suggest that employers with a higher number of employees may experience smoother or more successful visa processing, which could be a factor for applicants to consider.

5. **Strategic Continental Focus:** Investigate the specific reasons for lower certification rates from certain continents (e.g., Asia) to provide tailored advice, or strategically allocate resources to applicants from regions with historically higher approval rates.

6. **Leverage Tuned Models on Original Data:** For internal predictions and recommendations, continue using hyper-tuned ensemble models (specifically XGBoost and RandomForest) trained on the original, imbalanced dataset, as these demonstrated superior performance.

7. **Data Quality Check:** Address the negative values observed in no_of_employees in future data collection or preprocessing to ensure data integrity.

VI.3. Recommended Model

VI.3.1. Best Model Recommendation:

XGBoost is the recommended model for predicting visa `case_status`. It achieved the highest F1-score (0.824) and Recall (0.919) on the test set. The optimal parameters for this model are `subsample`: 0.8, `n_estimators`: 150, `max_depth`: 5, `learning_rate`: 0.01, and `colsample_bytree`: 0.8.

VI.3.2. Second Best Model Recommendation:

Tuned Decision Tree is the second-best recommended model for predicting visa case_status.

Test Set F1-score: 0.819 and **Test Set Recall:** 0.915 and the optimal parameters for this model are min_samples_split: 18, min_samples_leaf: 25, max_depth: 5, and criterion: 'entropy'. It showed a significant improvement after tuning and generalized well to the test data, achieving a very high recall.

VI.3.3. Third Best Model Recommendation:

The **Tuned RandomForest** model is the third-best recommended model.

Test Set F1-score: 0.817 and **Test Set Recall:** 0.891 and the optimal parameters for this model are n_estimators: 100, min_samples_split: 18, min_samples_leaf: 5, max_depth: 7, and criterion: 'entropy'. RandomForest consistently demonstrated strong performance and good generalization capabilities, offering a reliable alternative to XGBoost and the Decision Tree.

VII. Conclusion

The analysis conclusively demonstrates that educational attainment, job experience, and prevailing wage are the primary drivers for visa certification. By focusing on these key factors and utilizing robust ensemble models trained on the original dataset, EasyVisa can effectively predict visa outcomes and provide targeted, data-driven recommendations to both applicants and employers, optimizing the visa approval process.